

Polysaccharide breakdown products drive degradation-dispersal cycles of foraging bacteria through changes in metabolism and motility

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Abstract

Most of Earth's biomass is composed of polysaccharides. During biomass decomposition, polysaccharides are degraded by heterotrophic bacteria as a nutrient and energy source and are thereby partly remineralized into CO₂. As polysaccharides are heterogeneously distributed in nature, following the colonization and degradation of a polysaccharide hotspot the cells need to reach new polysaccharide hotspots. Even though many studies indicate that these degradation-dispersal cycles contribute to the carbon flow in marine systems, we know little about how cells alternate between polysaccharide degradation and motility, and which environmental factors trigger this behavioral switch. Here, we studied the growth of the marine bacterium *Vibrio cyclitrophicus* ZF270 on the abundant marine polysaccharide alginate, both in its soluble polymeric form as well as on its breakdown products. We used microfluidics coupled to time-lapse microscopy to analyze motility and growth of individual cells, and RNA sequencing to study associated changes in gene expression. We found that single cells grow at reduced rate on alginate until they form large groups that cooperatively break down the polymer. Exposing cell groups to digested alginate accelerates cell growth and changes the expression of genes involved in alginate degradation and catabolism, central metabolism, ribosomal biosynthesis, and transport. However, exposure to digested alginate also triggers cells to become motile and disperse from cell groups, proportionally increasing with the group size before the nutrient switch, and this is accompanied by high expression of genes involved in flagellar assembly, chemotaxis, and quorum sensing. We found that motile cells chemotax toward polymeric but not digested alginate, likely enabling them to find new polysaccharide hotspots. Overall, our findings reveal cellular mechanisms that might also underlie bacterial degradation-dispersal cycles, which influence the remineralization of biomass in marine environments.

Importance

Polysaccharides, also known as glycans, are the most abundant form of biomass on Earth and understanding how they are degraded by microorganisms is essential for our understanding of the global carbon cycle and the storage and release of CO₂ by natural systems. Although group formation is a common strategy used by bacterial cells to degrade ubiquitous polymeric growth substrates in nature, where nutrient hotspots are heterogeneously distributed, little is known about how cells prepare for dispersal from an exhausted nutrient source and re-initiate degradation of new nutrient patches. By quantifying growth, motility and chemotaxis of individual cells and comparing gene expression changes when populations were exposed to either polysaccharides or their degradation products in the form of digested polysaccharides, we show that bacterial cells alter their behavior when they experience a shift from polymeric to digested polysaccharides: After cells form groups during growth on polymers, the exposure to degradation products triggered cells to become motile, enabling dispersal from sessile cell groups. The chemotactic attraction of motile cells to polymeric alginate likely allows cells to move towards new polysaccharide hotspots. Our study sheds light on the cellular processes that drive bacterial growth and behavior during carbon remineralization, an important process resulting in CO₂ release from natural systems.

eLife assessment

This manuscript is a **valuable** contribution to our understanding of foraging behaviors in marine bacteria. The authors present a conceptual model for how a marine bacterial species consumes an abundant polysaccharide. Using experiments in microfluidic devices and through measurements of motility and gene expression, the authors offer **convincing** evidence that the degradation products of polysaccharide digestion can stimulate motility.

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Introduction

Polysaccharides represent the largest fraction of biomass on Earth^{1,2} and are constantly degraded and remineralized by microorganisms. Polysaccharides, also known as glycans, are long chains of monosaccharide units produced by cells for structural support (e.g., cellulose, chitin, and alginate) or energy storage (e.g., starch or glycogen)¹. Heterotrophic microbes obtain nutrients and energy from the polysaccharide breakdown products. They often use exoenzymes, either secreted or anchored in the cell membrane^{2,3}, to cleave these large polymers into smaller units that can be taken up by cells. The formation of dense cell groups is observed during growth on polysaccharides by diverse bacteria, including the soil- and gut-dwelling *Bacillus subtilis*³, the oligotrophic fresh-water *Caulobacter crescentus*⁴, and several representatives of the copiotrophic^{5,6} marine *Vibrio spp.*^{7,8}. It has been suggested that “cooperative” growth³, where dense cell groups reduce diffusional loss of valuable degradation products and secreted exoenzymes, represents a general principle in polysaccharide degradation^{4,8}. Yet, how bacteria subsequently disperse from exhausted polysaccharide sources and navigate towards new polysaccharide hotspots remains poorly understood.

To study the cellular mechanisms that govern the switch between polysaccharide degradation and dispersal toward new polysaccharide hotspots, we worked with the ubiquitous marine Gammaproteobacterium *Vibrio cyclitrophicus* ZF270, a degrader of the prevalent marine polysaccharide alginate^{6,9}. Alginate is a linear polysaccharide that is produced by brown algae as a cell wall component, as well as by certain bacteria. It can be cleaved into oligomers by endo-acting alginate lyases or into monomers by exo-acting alginate lyases (alginate lyases reviewed here¹⁰), namely β -D-mannuronic acid and α -L-guluronic acid¹¹. *V. cyclitrophicus* ZF270 is found predominantly in the large-particle fractions of coastal water¹², can attach and form biofilms on the surfaces of multiple polysaccharides including alginate¹³, and was shown to secrete alginate lyases during alginate degradation⁸. In this study, we used microfluidics coupled to automated time-lapse imaging to quantify the growth dynamics, group formation, and motility of *V. cyclitrophicus* ZF270 at the single-cell level under constant supply of either polymeric alginate or alginate degradation product in the form of digested alginate, as well as upon a transition from alginate to digested alginate. Furthermore, we used RNA-sequencing to compare the gene expression of *V. cyclitrophicus* ZF270 grown on alginate and digested alginate. We found striking responses to the form of alginate in growth rate, group formation, motility and chemotaxis, as well as in the expression of corresponding genes. Overall, our work provides insights into the metabolic and cellular regulation that allows cells to forage in heterogeneous nutrient-scapes through degradation-dispersal cycles.

Results

Extracellular break down of alginate delays population growth

To probe the phenotypic and metabolic regulation of bacterial cells during the progressive degradation of a polysaccharide source, we developed an experimental system consisting of the marine bacterium *V. cyclitrophicus* ZF270 growing on alginate, a highly abundant polysaccharide in the ocean. To mimic the local nutrient environment during the colonization of a new polysaccharide source, we used 0.1% weight per volume (w/v) algae-derived alginate in its soluble form (in the following also simply referred to as “alginate”). Since commercially available alginate breakdown products of specific sizes are limited and expensive, with only one monomeric component available at a considerable cost, we simulate an advanced stage of polysaccharide degradation by supplying cells with digested alginate (0.1% (w/v)). This digested alginate is prepared by treating alginate with a readily available endo-acting alginate lyase (see Materials and Methods). This mimics an environment where degradation products like monomers and oligomers become abundant through the action of extracellular alginate lyases. The commercially available alginate lyase has been shown to produce alginate oligomers of progressively smaller size over extended digestion periods¹⁴. We used Liquid Chromatography-Mass Spectrometry (LC-MS) to analyze the composition of the digested alginate, which had been subject to 48 hours of digestion. We found that digested alginate contained more monosaccharides than untreated alginate, which contained more alginate molecules of higher molecular weight (Fig. S1).

Using this system, we first set out to investigate the growth dynamics of *V. cyclitrophicus* ZF270 in well-mixed batch cultures containing either alginate or digested alginate as a sole carbon source (Fig. S2A). Measurements of optical density showed that the onset of growth on alginate was delayed by about 7.5 hours compared to growth on digested alginate (Fig. S2B, D and E). This is consistent with previous observations that in well-mixed environments the lag time of bacteria growing on alginate can be reduced by the external supplementation of alginate lyases⁸. We also observed that the optical density at stationary phase was higher when cells were grown on alginate (Fig. S2B and C). However, colony counts did not show a significant difference in cell numbers (Fig. S3), suggesting that the increased optical density may stem from aggregation of cells in the alginate medium, as observed for other *Vibrio* species⁷. Overall, these findings indicate

that growth on polysaccharides such as alginate in well-mixed cultures is initially limited by the extracellular breakdown of the polysaccharide, though similar cell numbers were reached eventually.

Large cell groups form on alginate but not on digested alginate

To better understand how cells react to the changing degree of depolymerization of a polysaccharide source during degradation, we investigated the growth of *V. cyclitrophicus* ZF270 at the level of single cells on alginate and digested alginate. For this purpose, we grew the cells in microfluidic growth chambers, as depicted in **Figure 1A** and described in detail by Dal Co et al.¹⁵ In brief, the microfluidic chips are made of an inert polymer (polydimethylsiloxane) bound to a glass coverslip. The PDMS layer contains flow channels through which the culture medium is pumped continuously. Each channel is connected to several growth chambers that are laterally positioned. The dimensions of these growth chambers (height: 0.85 μm , length: 60 μm , width: 90–120 μm) allow cells to freely move and grow as monolayers. The culture medium, containing either alginate or digested alginate in their soluble form, is constantly pumped through the flow channel and enters the growth chambers primarily through diffusion.^{15,16,4,17,8} Therefore, the number of cells and their positioning within microfluidic chambers is determined by the cellular growth rate as well as by cell movement.⁴ This setup combined with time-lapse microscopy allowed us to follow the development of cell communities over time. We found that over 24 hours dense groups of more than 1000 cells formed on alginate, filling the entire microfluidic chamber (**Fig. 1B**). In contrast, cells supplied with digested alginate grew in smaller groups that never exceeded 100 cells per chamber (**Fig. 1C and D**). Reconstructed cell lineages revealed that the large cell groups on alginate formed because cells often did not disperse after division and thereby formed dense cell groups originating from a single cell lineage (**Fig. 1B**). The lower cell density on digested alginate could be caused by slower growth or by more cells leaving the chambers. As the maximum growth rate both in bulk and on single cell level is similar in alginate and digested alginate (**Fig. S2E and F**), it is more likely that the lower cell density in digested alginate is caused by cell dispersal. To test the role of increased viscosity of polymeric alginate in causing the increased aggregation of cells, we measured the viscosity of 0.1% (w/v) alginate or digested alginate dissolved in TR media. For alginate, the viscosity was 1.03 ± 0.01 mPa·s (mean and standard deviation of three technical replicates) whereas the viscosity of digested alginate in TR media was found to be 0.74 ± 0.01 mPa·s. Both these values are relatively close to the viscosity of water at this temperature (0.89 mPa·s¹⁸) and, while they may affect swimming behavior,¹⁹ they are insufficient to physically restrain cell movement.²⁰ Overall, our observations suggest that cells can modulate their propensity to form groups depending on the state of polysaccharide degradation in their local environment. Similar observations were made with a different model system (*Caulobacter crescentus* growing on the polysaccharide xylan)¹⁶, indicating that group formation on polymeric nutrient sources may be a general mechanism of bacteria that degrade polysaccharides extracellularly.

Transition from alginate to digested alginate triggers density-dependent dispersal of cells

To investigate the transition from the large cell groups formed on alginate to the small groups formed on digested alginate, we subjected *V. cyclitrophicus* ZF270 cells grown on alginate to a switch to digested alginate (**Fig. 2A**). Following the limited cell motility on alginate, this switch led to a rapid decrease in cell density within the growth chambers (**Fig. 2B**), presumably caused by cell dispersal. As we previously reported for other *Vibrionaceae* isolates, the growth rate of the cells on alginate was dependent on the local cell density: Initially the growth rate increased with cell density but then decreased at high cell densities, indicating that cell groups can benefit from the sharing of breakdown products generated by each other's exoenzymes, but also increasingly compete for nutrients.⁸ This led us to investigate whether cells in larger groups, potentially experiencing stronger nutrient competition, might have a higher propensity to disperse after a

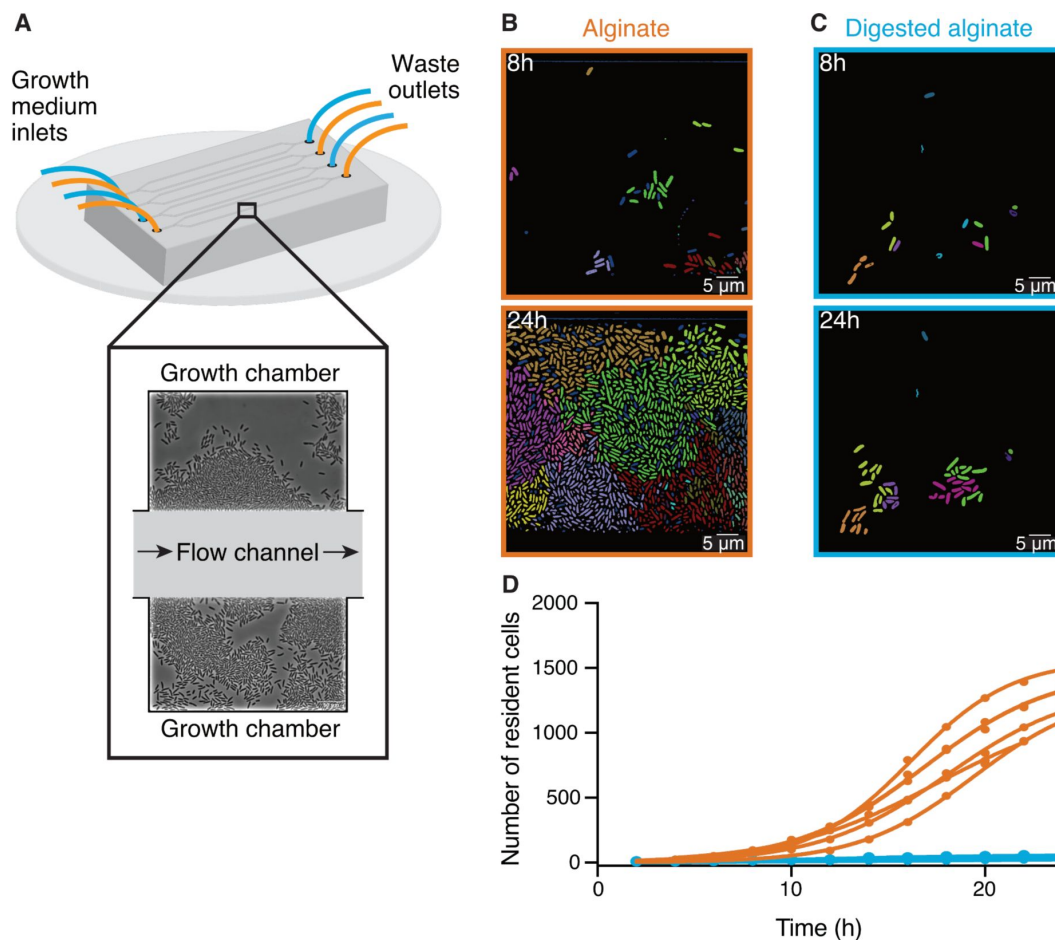


Figure 1.

Large cell groups form on alginate but not on digested alginate.

(A) Schematic representation of the setup of the microfluidic experiments. (B and C) Representative images at different time points of *V. cyclitrophicus* ZF270 cells growing in microfluidic chambers, described in detail by Dal Co et al.¹⁵, with (B) alginate medium or (C) digested alginate medium, both in their soluble form (not visible). Cells are false-colored according to their lineage identities based on cell segmentation and tracking over 24 hours. Cells without identified progenitors are colored in dark blue. See Supplemental Videos S1 (alginate) and S2 (digested alginate) for time-lapse videos. (D) Cell numbers within microfluidic chambers supplied with alginate (orange) are substantially higher than cell numbers within microfluidic chambers supplied with digested alginate (blue) (Logistic growth regression for alginate: $R^2 = 0.99$, maximal number of cells = 1217-1564, $k = 0.24$ - 0.38 h^{-1} ; for digested alginate: $R^2 = 0.86$ - 0.97 , maximal number of cells = 47-100, $k = 0.07$ - 0.4 h^{-1}). Circles indicate the number of cells present at a given time point in each chamber ($n_{\text{chambers}} = 7$). Data for chambers with alginate originate from D'Souza et al.⁸. Lines are fits of a logistic growth regression line for each condition.

switch to digested alginate than cells in smaller groups. We indeed found that the nutrient switch caused a few or no cells to disperse from small cell groups (**Fig. 2B** [↗](#)), whereas a large fraction of cells from large cell groups dispersed (**Fig. 2C** [↗](#)). In fact, the fraction of cells that dispersed upon imposition of the nutrient switch showed a strong positive relationship with the number of cells present, meaning that cells in chambers with many cells were more likely to disperse than cells in chambers with fewer cells (**Fig. 2C** [↗](#)). Thus, during the transition from polysaccharides to degradation products, we found that the dispersal rate of cells depends on the size of the cell groups, likely through increased motility of cells in large groups.

Cells growing on digested alginate are more motile and polymeric alginate acts as chemoattractant

To investigate whether increased cell motility of *V. cyclitrophicus* ZF270 underlies the dispersal of cells observed after a shift to digested alginate, as also previously observed for *C. crescentus* cells on the monosaccharide xylose^{4,16} [↗](#), we quantified the motility of single cells supplied with either alginate or digested alginate. We found that the single-cell swimming speed as well as the swimming distance were significantly larger for cells supplied with digested alginate compared to cells supplied with alginate, and that a larger fraction of cells was motile (**Fig. 3** [↗](#)). This confirmed increased motility as a cellular response to the exposure to degradation products in the form of digested alginate. Overall, these findings suggest that once the breakdown of a polysaccharide source makes breakdown products available in the local environment, a fraction of cells becomes motile and disperses.

To understand whether alginate polymers or alginate breakdown products act as chemoattractants on motile cells, we measured the chemotactic strength towards alginate and digested alginate using the In Situ Chemotaxis Assay (ISCA)^{21,22} [↗](#). Interestingly, we found *V. cyclitrophicus* ZF270 to significantly chemotax toward alginate (chemotactic index $I_c > 1$) but not significantly toward digested alginate (**Fig. 3F** [↗](#)). This suggested that the increase in motility is accompanied by chemotaxis toward alginate polymers.

Altered gene expression in central carbon metabolism, enzyme production, secretion and transporters, motility, and quorum sensing underlies the late-stage alginate degradation and cell dispersal

Next, we sought to elucidate the molecular mechanisms underlying the observed phenomenological disparities between cells cultivated on alginate and digested alginate. Due to the challenge of generating knock-out mutants in natural isolates, we used transcriptomics to investigate the differentially expressed genes of *V. cyclitrophicus* ZF270 under these respective conditions. To obtain a high-quality reference genome of *V. cyclitrophicus* ZF270, we sequenced and assembled a new reference genome using combined short and long read sequencing (BioProject PRJNA991487). We then grew cultures on either alginate or digested alginate until mid-exponential phase. To understand which cellular functions were affected by the expression changes, we first performed differential gene expression analysis using the software DESeq2. Here, genes exhibiting a log2 fold expression change greater than 0.5 or smaller than -0.5 between the two conditions, with a Benjamini-Hochberg(BH)-adjusted *p*-value below 0.01, were considered to be differentially expressed (Table S1). Next, we investigated which KEGG categories were enriched in either genes with increased or decreased expression via Gene Set Enrichment Analysis (GSEA)²³ [↗](#) (Table S2) and found nine categories significantly enriched in genes with increased gene expression on digested alginate, and three categories significantly enriched in genes with decreased gene expression (**Fig. 4A** [↗](#)).

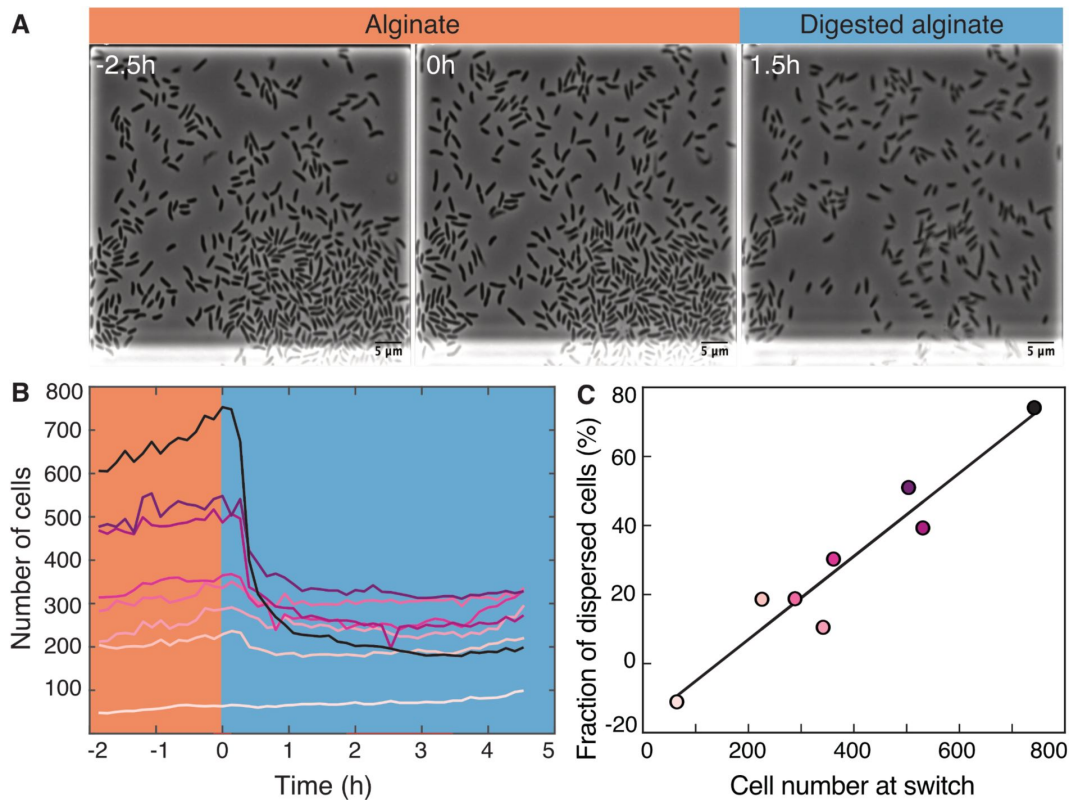


Figure 2.

Transition from alginate to digested alginate triggers density-dependent dispersal of cells.

(A) Representative time-lapse images of *V. cyclitrophicus* ZF270 cells (phase contrast microscopy) in microfluidic growth chambers that were initially exposed to alginate and then switched to digested alginate. (B) Number of cells in different chambers over time, each chamber indicated by a unique color ($n = 8$). The carbon source is indicated by the colored background (orange: alginate; blue: digested alginate). See Supplemental Video 3 for a time-lapse video. (C) Positive relationship between the number of cells in the microfluidic growth chamber at the time of the switch and the fraction of cells that disperse after the switch. Each circle represents one growth chamber with colors corresponding to (B), and the line depicts a linear regression fit ($R^2 = 0.92$, slope = 0.12).

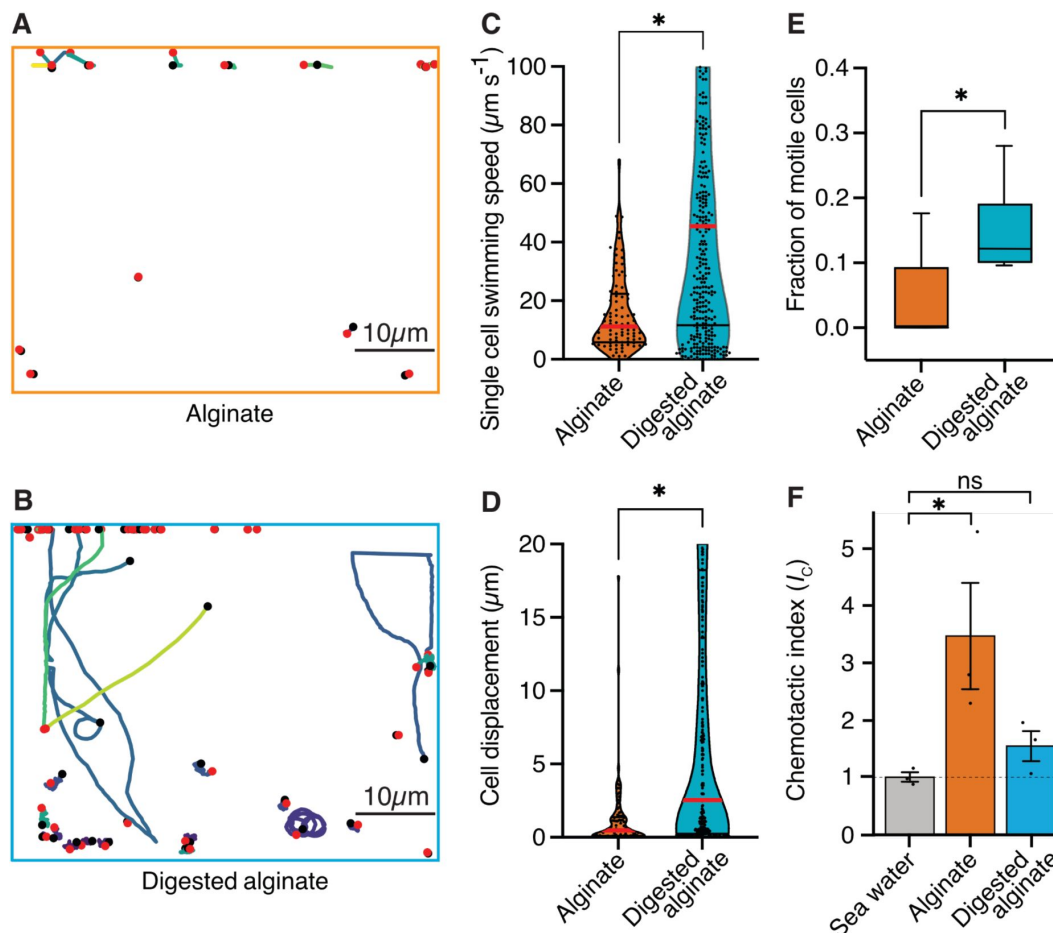


Figure 3.

Cells are more motile on digested alginate than alginate and show chemotaxis towards alginate.

(A and B) Spatial trajectories of cells supplied with (A) alginate or (B) digested alginate in representative microfluidic growth chambers are shown. Black points mark the starting point of each trajectory, pink points mark the end point of each trajectory, and colored lines mark the trajectories of cells. (C) Distributions of the mean single-cell swimming speeds (Nested t-test, p -value < 0.0007, $t = 4.803$, $\text{df} = 10$, $n_{\text{cells}} = 86$ vs 375 , and $n_{\text{chambers}} = 5$) are shown. (D) Distributions of cell displacement over the course of a trajectory (Nested t-test test, p -value < 0.0131, $t = 4.39$, $\text{df} = 10$, $n_{\text{cells}} = 86$ vs 375 , and $n_{\text{chambers}} = 5$) are shown. In (C) and (D) the red horizontal lines indicate the mean while black lines depict the 25th and 75th quartiles of the distribution. (E) The mean fraction of motile cells in each chamber, where motile cells are defined as cells with displacement greater than $1 \mu\text{m}$ (Mann-Whitney test on the means of five growth chambers, p -value = 0.034). In C, D, and E, each chamber was considered as an independent replicate. (F) Chemotactic index (I_C) quantified by In Situ Chemotaxis Assay (ISCA) (Tukey multiple comparisons of means, 95% family-wise confidence levels as error bars, p -value < 0.05, $n = 3$). Asterisks indicate statistically significant differences. See Supplemental Videos S4 and S5 for time-lapse videos of swimming cells.

Cells growing on digested alginate showed expression changes in parts of the central metabolism and translation, compared to cells growing on alginate. Specifically, two pathways of the central metabolism were significantly enriched in genes with lower expression on digested alginate (“Valine, leucine and isoleucine biosynthesis” and “Propanoate metabolism” with a negative normalized enrichment score (NES), **Fig. 4A** [↗](#), S4A and B, Table S3 and S4). We also found the gene set that encodes ribosomal proteins significantly enriched with genes highly expressed on digested alginate (positive NES, **Fig. 4A** [↗](#) and S4C, Table S5). This implies that cells invest proportionally more of their transcriptome into the production of new proteins, a sign of faster growth^{24,25}. Both observations likely relate to the different growth dynamics of *V. cyclitrophicus* ZF270 on digested alginate compared to alginate (Fig. S2A), where cells in digested alginate medium reached their maximal growth rate 7.5 hours earlier and thus showed a shorter lag time (Fig. S2D). As a consequence, the growth rate at the time of RNA extraction (mid-to-late exponential phase) may have differed, even though the maximum growth rate of cells grown in alginate medium and digested alginate medium were not found to be significantly different (Fig. S2E).

The expression of transporters and secretion systems was generally increased in cells growing on digested alginate, compared to cells growing on alginate. This includes genes with a function in “Bacterial secretion systems”, namely secretion systems I to VI, which mediate protein export through the inner and outer membranes of Gram-negative bacteria (**Fig. 4A** [↗](#) and S4D, Table S6). Notably, eleven of the thirteen General Secretion Pathway (GSP) genes (part of the type II secretion system) showed 1.1 to 3.4-fold increased expression levels on digested alginate (*p*-values < 0.01, **Fig. 4A** [↗](#), Table S8). The GSP is known to facilitate secretion of various extracellular enzymes like chitinases, proteases, and lipases^{26,27}, therefore, the positive enrichment on digested alginate may be linked to the export of extracellular alginate lyases. The KEGG BRITE category “Secretion system” was also enriched on digested alginate (**Fig. 4A** [↗](#) and S4D, Table S7). It contains additional genes involved in protein export, implicating them as interesting subjects to further research in their role in protein secretion and cell attachment and detachment during degradation-dispersal cycles of polysaccharide-degrading bacteria. Also, the gene set of transporters and in particular ABC transporters were positively enriched (**Fig. 4A** [↗](#), Table S10 and S11). The latter showed enrichment especially in genes related to saccharide, iron, zinc, and phosphate transport (Fig. S4E), which suggests that cells growing on digested alginate invest proportionally more of their transcriptome into uptake of not only saccharides but also essential nutrients like iron, zinc, and phosphate, which may become growth-limiting when degradation products are abundantly available.

Both motility and quorum sensing genes increased in expression in cells growing on digested alginate. The positive enrichment of the gene set containing bacterial motility proteins aligned with our expectations based on the increase in motile cells that we observed in **Figure 3E** [↗](#) (**Fig. 4A** [↗](#), Table S12). The set of quorum sensing genes was also positively enriched in cells growing on digested alginate (**Fig. 4A** [↗](#) and S4F, Table S13). This role in dispersal is in agreement with a previous study that showed induction of the quorum sensing master regulator in *V. cholerae* cells during dispersal from biofilms on a similar time scale as here (less than an hour)²⁸. Quorum sensing is known to control biofilm formation in the well studied model system *Vibrio cholerae*^{29,30} and may also orchestrate the density-dependent dispersal in the presence of degradation products observed in our study, though the particular signaling cues remain to be uncovered. The strong cellular response to the degree of alginate depolymerization was also emphasized by the finding that transcription factor genes were enriched among genes with decreased expression on digested alginate (**Fig. 4A** [↗](#), Table S14). The set of genes associated to KEGG’s beta-lactam resistance category was enriched in genes with increased expression on digested alginate. However, most genes in this gene set are also associated to the KEGG pathways “Transporters”, “Quorum sensing”, “Chromosome and associated genes”, and “Peptidoglycan biosynthesis” and thus the enrichment likely reflects expression changes in these categories and may relate to the difference in growth dynamics (**Fig. 4A** [↗](#) and S4G, Table S15). Overall, the

observed gene expression changes provide insights into the molecular mechanisms underlying the substantial adaptations of *V. cyclitrophicus* ZF270 cells to the polymeric and digested form of alginate.

Growth on digested alginate is associated with increased expression of alginate catabolism, flagellar motility and chemotaxis genes

For a more fine-grained understanding of the impact of the form of alginate on the alginate catabolism, we investigated specifically the expression of genes involved in alginate degradation, uptake, and catabolism. In the closed genome of *V. cyclitrophicus* ZF270, we identified genes that encode CAZymes responsible for alginate degradation, namely alginate lyase genes from the PL6, PL7, PL15, and PL17 family. We identified homologs of alginate transporters (porin *kdgM*, symporter *toaA*, *toaB*, and *toaC*) and metabolic enzymes that shunt into the Entner-Doudoroff pathway (DEHU reductase genes *dehR*, *kdgK*, and *eda*), based on the known genes in the alginate degradation pathway of *Vibrionaceae*^{31,32}. We found that the expression of most of these genes increased significantly on digested alginate relative to alginate (Fig. 4B, Fig. 5A, S8A, Table S16). Surprisingly, also the expression of most alginate lyase genes increased on digested alginate, especially the expression of secreted alginate lyase genes. This indicates the production of “public” exoenzymes despite the abundance of monomeric and oligomeric degradation products in the digested alginate medium.

The increased motility of cells observed upon exposure to digested alginate (Fig. 3, Fig. 4A) led us to evaluate the expression of motility- and chemotaxis-associated genes across digested alginate and alginate treatments. As flagella are the main mode of motility in the genus *Vibrio*^{33,34}, we focused on the expression of flagella-related genes. In the genome of *V. cyclitrophicus* ZF270 we found a gene cluster that encodes most genes involved in flagellar assembly and that was flanked by chemotaxis genes (hereon called flagellar locus, Fig. S8B, and Table S17). Overall, 21/34 flagellar locus genes were differentially expressed (log2 fold change > 0.5, BH-adjusted *p*-value < 0.01) and the majority (90%) of these differentially expressed genes showed increased expression on digested alginate (Fig. 4B). The flagellar genes *flgA*, *fljC*, and *fliH* and the chemotaxis gene *cheW* showed the strongest overexpression (5.4, 3.8, 2.0, and 2.6-fold, respectively). The expression of flagellar biosynthesis genes in *Vibrionaceae* occurs by a cascade of gene expression of four classes of genes (Class I - IV)³⁵. We found the expression of the master regulator of the flagellar biosynthesis regulon, the Class I gene *flrA*, to be 1.6-fold increased on digested alginate (BH-adj. *p*-value 3e-19)³⁶. The Class II regulatory genes *flrBC*, controlling Class III genes, and *fliA*, controlling Class IV genes, showed 1.5-fold and 1.3-fold increased expression on digested alginate (BH-adj. *p*-value 1e-12 for *flrB*, 2e-39 for *flrC*, 6e-15 for *fliA*) (Fig. 5B)^{34,37}. These findings suggest that the increased phenotypic motility observed on digested alginate (Fig. 3) is related to the upregulation of flagellar biosynthesis genes. Additionally, the expression of the flagellum filament, encoded by *fliC* genes of the flagellar locus, was partially increased: *flaD* and *flaC* expression were increased by 3.8 and 1.6-fold, whereas the *flaA* gene was not significantly differentially expressed (BH-adj. *p*-value 7e-37, 2e-19, and 1e-02, respectively) (Fig. 5B, Table S6). This suggests that cells grown on digested alginate have a *flaD*-rich filament composition, which has been shown to alter the swimming and adhesion characteristics of bacterial cells^{38,39}. Lastly, we found that most genes involved in chemotaxis and located in the flagellar locus are highly expressed in cells grown on digested alginate (Fig. 5C, Fig. 4B) and likely drive the chemotactic activity of *V. cyclitrophicus* ZF270 during dispersal. Overall, our findings elucidate that cellular responses upon exposure to degradation products manifest in increased expression of genes involved in extracellular alginate breakdown and alginate catabolism as well as flagellar assembly and chemotaxis.

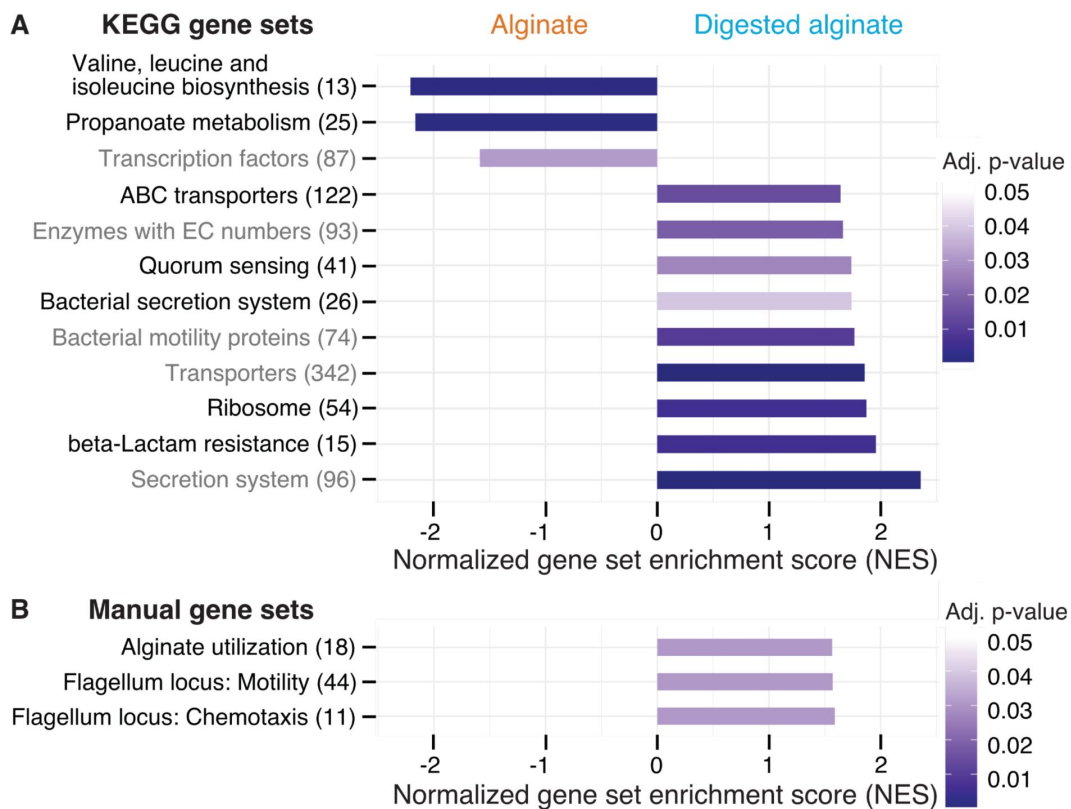


Figure 4.

Twelve functional gene sets are enriched in genes with increased or decreased expression in cells grown on digested alginate.

Gene set enrichment analysis (GSEA) with (A) all KEGG pathways and KEGG BRITE categories as gene sets or with (B) a custom alginate utilization, flagellar assembly, and flagellum-driven chemotaxis gene set was performed comparing the gene counts of the transcriptome of *V. cyclitrophicus* ZF270 cultures grown on digested alginate and alginate (six replicates each). Gene sets with a positive enrichment score were enriched with genes with higher expression in cells grown on digested alginate relative to cells grown on alginate (BH-adjusted p -value < 0.05), whereas gene sets with negative enrichment scores were significantly enriched with genes with decreased expression on digested alginate. The number in brackets indicates the number of genes with unique K number per gene set (A) and the number of genes per gene set (B) within the *V. cyclitrophicus* ZF270 genome.

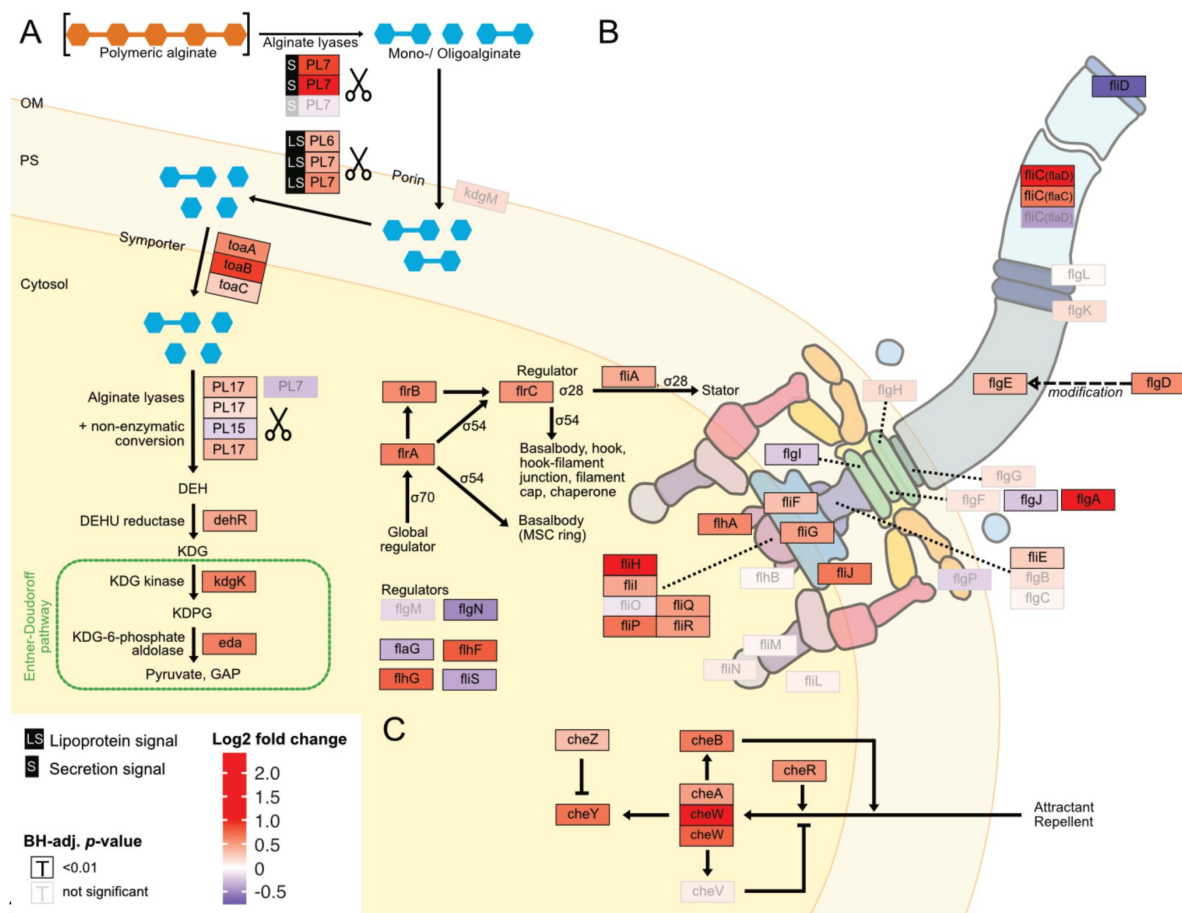


Figure 5.

Digested alginate increases expression of genes involved in alginate degradation, uptake and catabolism, as well as flagellar assembly and chemotaxis.

Genome-wide differential expression analysis where the log₂ fold changes of gene expression on digested alginate compared to alginate is shown for (A) alginate lyases (PL6, PL7, PL15, PL17, scissors symbol), transporters (porin *kdgM*, symporter *toaB*, symporter *toaC*), and metabolic enzymes shunting into the Entner-Doudoroff pathway (DEHU reductase *DehR*, *kdgK*, *eda*), (B) genes of the flagellar locus associated with flagellar assembly and (C) adjacent chemotaxis genes. Genes displayed in (B) and (C) are part of the KEGG pathways “Bacterial motility proteins” and “Bacterial chemotaxis”. Differential expression analysis was performed to compute the Benjamini-Hochberg-adjusted Wald test *p*-value (“BH-adj. *p*-value”, text color and box outline color) and log₂ fold change (box fill color) for each gene (box). For better visibility, genes that exhibited a log₂ fold gene expression change greater than 1 (i.e., doubling of expression) or less than -1 (i.e., halving of expression) are designated maximum intensity of red or blue, respectively. Genes with BH-adj. *p*-value smaller than 0.01 were considered significantly differentially expressed. In (A), the location of the gene products was based on [Figure 1](#) of Wargacki et al.³¹ with the exception of the alginate lyases (PL6, PL7, PL15, PL17) which were placed based on their signal peptides (S: extracellular, LS: membrane-embedded, none: cytosolic). In (B) and (C) the gene location and depiction were based on the KEGG pathway “Flagellar assembly” (map02040), “Bacterial chemotaxis” (map02030), and [Figure 3](#) of Rajagopala et al.⁴⁰. Genes without known cellular location were omitted here but displayed in the genomic architecture in Fig. S5. Arrow: activation; dashed arrow: modification; “flat” arrow: inhibition; OM: outer membrane; PM: periplasm; IM: inner membrane; PL: polysaccharide lyase family; *kdgM*: oligogalacturonate-specific outer membrane porin; *toaABC*: oligoalginate symporter; DEH: 4-deoxy-L-erythro-5-hexoseulose uronic acid; *dehR*: DEH reductase; KDG: 2-keto-3-deoxy-gluconate; *kdgK*: KDG kinase; KDPG: 2-keto-3-deoxy-6-phosphogluconate; *eda*: KDG-6-phosphate aldolase; GAP: glyceraldehyde 3-phosphate; ED: Entner-Doudoroff; ns: not significant, i.e. BH-adj. *p*-value > 0.01.

Discussion

On Earth organic carbon is mostly present in the form of polysaccharides^{1,2}, which are often in a particulate state and form a heterogeneous resource landscape. Over the last years, the study of extracellular bacterial degradation of polysaccharides has revealed that bacterial growth on polysaccharides increases with increased cell density, enabling cells to benefit from the exoenzymes and extracellular degradation products of surrounding cells otherwise lost to diffusion (“cooperative growth”)^{41–43}. However, cells have been observed not only to aggregate on polysaccharide sources, but also to leave them before the source is depleted^{13,43}. This cycle of biomass degradation and dispersal has long been discussed in the context of foraging e.g.,^{44,45,13,46,47}, but the cellular mechanisms that drive the cell dispersal remain unclear.

Our work links these observations and connects them to the cellular mechanisms that underlie the degradation-dispersal cycles of bacterial degraders, which we see as basal drivers of the biogeochemical processing of polysaccharides in heterogeneous nutrient-scapes. When bacteria encounter a new source of biomass, their local environment likely contains few mono- or oligosaccharides but is rich in polysaccharides which usually require extracellular breakdown (**Fig. 6**, “Finding a new nutrient source”). General concepts of how bacteria recognize the presence of large biopolymers remain elusive, but it was proposed that “sentry” enzymes are constitutively expressed at a basal level to cleave mono- or oligosaccharides from polysaccharides, which cells can take up and which prime their metabolism for the degradation of the respective polysaccharide^{48,49}. It is not known yet how widespread the concept of sentry enzymes may be, but the observation of a constitutively expressed PL7 and PL15 family alginate lyase gene in *Z. galactanivorans*⁴⁸ is mirrored in our work by the constant expression of an extracellular PL7 family alginate lyase gene, which may act as sentry enzyme that helps to initiate alginate degradation when cells encounter alginate.

Growth on polysaccharides has been found to be dependent on the cell density, as increased cell density limits the loss of degradation products and exoenzymes by diffusion⁸ (**Fig. 6**, “Early-stage polysaccharide breakdown”). We found that cells grown on degradation products reach their maximal growth earlier and show increased expression of ribosomal biosynthesis, enzyme secretion, especially of secreted alginate lyases, transporters, quorum sensing and expression changes in the central carbon metabolism. The secretion of alginate lyases might seem surprising and wasteful in a monomer-rich environment. One reason for this observation may be that cells primarily rely on intracellular monosaccharide levels to trigger the upregulation of genes associated with polysaccharide degradation and catabolism, as has previously been observed for *E. coli* across various carbon sources^{50,51}. In fact, the majority of carbon sources are sensed by prokaryotes through one-component sensors inside the cell⁵⁰. In the one-component internal sensing scheme, the enzymes and transporters for the use of various carbon sources are expressed at basal levels, which leads to an increase in pathway intermediates upon nutrient availability. The pathway intermediates are sensed by an internal sensor, usually a transcription factor, and lead to the upregulation of transporter and enzyme expression^{50,51}. This results in a positive feedback loop, which enables small changes in substrate abundance to trigger large transcriptional responses^{50,52}. Thus, the presence of alginate breakdown products may likely result in increased expression of all components of the alginate degradation pathway, including the expression of degrading enzymes. As the gene expression analysis was performed on well-mixed cultures in culture medium containing alginate breakdown products, we therefore expect a strong stimulation of alginate catabolism. In a natural scenario, where cells disperse from a polysaccharide hotspot before its exhaustion, the expression of alginate catabolism genes may likely decrease again once the local concentration of breakdown products decreases. However, continued production of alginate lyases could also provide an advantage when encountering a

new alginate source and continued production of alginate lyases may thus help cells to prepare for likely future environments. Further investigations of bacterial enzyme secretion in changing nutrient environments and at relevant spatial scales are required to improve our understanding of the regulation of enzyme secretion along nutrient gradients.

We show that cells respond to the exposure to degradation products with dispersal from dense cell groups by means of increased flagellum-driven swimming (**Fig. 6**, “Late-stage polysaccharide breakdown”), decreasing their local cell number. This finding matches with previous observations of cells leaving biopolymer particles before they are depleted^{13,43}. A plausible explanation for this density-dependent dispersal is that cells in larger groups compete with each other for nutrients and space, while not profiting from cooperative degradation anymore due to the abundance of degradation products. Motility has also been shown to increase the encounter rate of cells with sources of nutrients^{53,54}, suggesting motility as a strategy that allows cells to escape from the ensuing competition. Previous work in *Caulobacter crescentus* demonstrated that a flagellum knock-out mutant formed larger cell groups, resulting in reduced growth rates due to intercellular competition^{4,16}. While it would be interesting to study non-motile mutants of *V. cyclitrophicus* ZF270, the non-tractability of natural isolates makes direct tests of molecular mechanisms difficult. Additionally, subjecting cells separately to the two monomeric units of alginate or oligomers of defined size could improve our understanding of the specific molecules that trigger motility, but this was experimentally not feasible.

Direct chemotaxis towards polysaccharides may facilitate the search for new polysaccharide sources after dispersal. We found that the presence of degradation products not only induces cell dispersal but also increases the expression of chemotaxis genes. Interestingly, we found that *V. cyclitrophicus* ZF270 cells show chemotaxis towards polymeric alginate but not digested alginate. This contrasts with previous findings for bacterial strains degrading the insoluble marine polysaccharide chitin, where chemotaxis was strongest towards chitin oligomers⁵³, suggesting that oligomers may act as an environmental cue for polysaccharide nutrient sources⁵⁵. However, recent work has shown that certain marine bacteria are attracted to the marine polysaccharide laminarin, and not laminarin oligomers⁵⁶. Together with our results, this indicates that chemotaxis towards soluble polysaccharides may be mediated by the polysaccharide molecules themselves. The mechanism of this behavior is yet to be identified, but could be mediated by polysaccharide-binding proteins as have been found in *Shingomonas* sp. A1 facilitating chemotaxis towards pectin⁵⁷. Direct polysaccharide sensing adds complexity to chemosensing as polysaccharides cannot freely diffuse into the periplasm, which can lead to a trade-off between chemosensing and uptake⁵⁸. Furthermore, most polysaccharides are not immediately metabolically accessible as they require degradation. But direct polysaccharide sensing can also provide certain benefits compared to using oligomers as sensory cues. First, it could enable bacterial strains to preferably navigate to polysaccharide nutrients sources that are relatively uncolonized and hence show little degradation activity. Second, strong chemotaxis towards degradation products could hinder a timely dispersal process as the dispersal then requires cells to travel against a strong attractant gradient formed by the degradation products. Overall, this strategy allows cells to alternate between degradation and dispersal to acquire carbon and energy in a heterogeneous world with nutrient hotspots^{44,59–61}.

Conclusion

The heterogeneous landscape of polysaccharide hotspots in natural systems requires bacteria to effectively break down polymeric carbohydrates as well as readily ensure dispersal to new nutrient hotspots. Our findings show that the degree of depolymerization of the polysaccharide influences this decision, altering the growth dynamics, metabolic activity, and motility of cells. Our study also contextualizes the surprising finding that foraging bacteria majorly leave polysaccharide particles before the last third of the particle is consumed⁴³. Dispersal from a

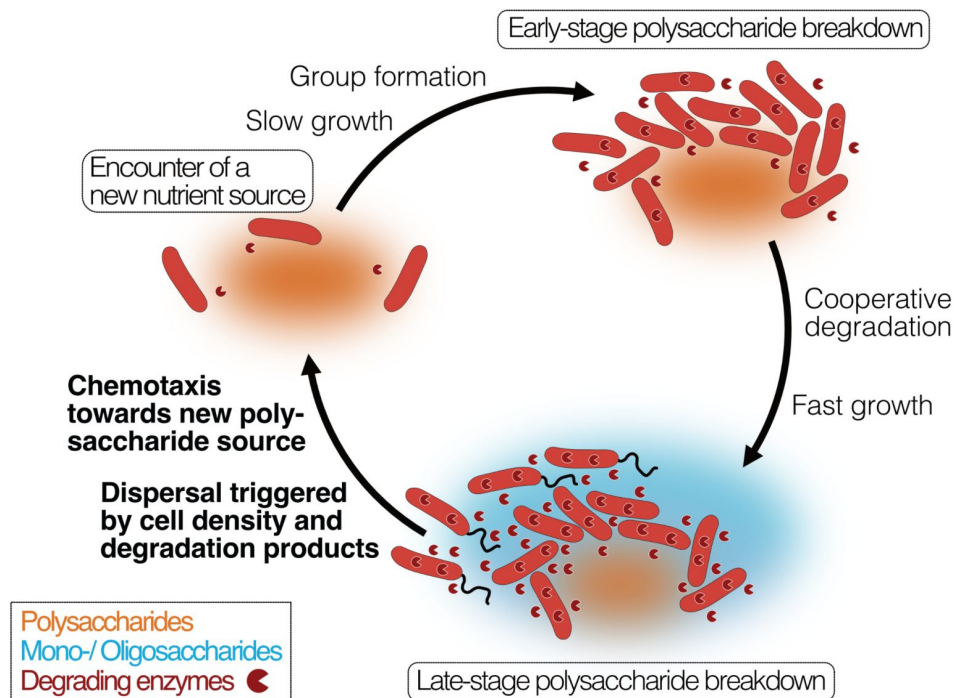


Figure 6.

Bacterial growth and regulation on patches of polysaccharides.

By integrating our results with previous studies on cooperative growth on the same system, as well as results on dispersal cycles in other systems, we highlight where the specific results of this work add to this framework (**bold font**). When cells encounter polymer sources, the colonization of the nutrient hotspot may be aided by the basal exoenzyme production of “sentry” enzymes (“Encounter of a new carbon source”). This phase is succeeded by group formation, which enables cells to benefit from exoenzymes of neighboring cells and diffusing degradation products (“Early-stage polysaccharide breakdown”). The following phase includes cooperative extracellular degradation of the polysaccharide source, further increasing the concentration of available degradation products. These degradation products trigger the overexpression of alginate degrading, importing, and catabolizing enzymes, ensuring swift polysaccharide degradation (“Late-stage polysaccharide breakdown”). The increased pool of breakdown products also cues flagellar swimming in a subpopulation of cells and increases the expression of chemotaxis genes. Polymeric alginate acts as chemoattractant towards new polysaccharide sources. Cells and molecules are not drawn to scale. Dark red pie symbols: intracellular and extracellular polysaccharide-degrading enzymes; orange shading: a polymeric carbon source; blue shading: monomeric or oligomeric degradation products.

partially degraded carbon source may serve several purposes: (i) escaping competition that ensues within large cell groups, (ii) ensuring the spread of a part of the clonal population to new environments as bet hedging strategy^{47,62}, here guided by chemotaxis towards new nutrient hotspots, (iii) preventing whole populations degrading a sinking marine particle or a deposited sediment particle to be buried in depth where nutrient hotspots become sparse^{43,63}, and/or (iv) increase the genetic variation in bacterial populations⁴⁷. However, dispersal may also occur when a nutrient source offers a surplus of carbon while other essential nutrients become limiting, as the increased expression of iron, zinc, and phosphate transporters in cells grown on digested alginate suggested. These findings emphasize that metabolic molecules can also act as triggers of dispersal, expanding upon the current perspective of dispersal in biofilms as a reaction to dispersal cues like NO, signaling molecules, nutrient starvation, and oxygen starvation⁶⁴. The study of bacterial motility on increasingly complex biomass particles will reveal the role of the nutrient composition of the present nutrient hotspot on the bacterial decision-making. Overall, these new insights into the cellular mechanisms and regulation that drive degradation-dispersal cycles contribute to our understanding of the microbially driven remineralization of biomass, and factors that modulate this process. The open questions of how bacteria sense polysaccharides in their environment, which cell signaling pathways integrate the presence of degradation products in the cellular decision-making of degradation and dispersal, and to what extent cell populations coordinate this decision, present an exciting avenue of further research.

Materials and Methods

Bacterial strains, media and growth assays

Vibrio cyclitrophicus ZF270 (available through Culture Collection Of Switzerland; Accession number: 2043) cells were cultured in Marine Broth (DIFCO) and grown for 18 hours at 25 °C. Cells from these cultures were used for growth experiments in Tibbles Rawling (TR) salts minimal medium^{65,66} containing either 0.1% (weight/volume) algae-derived alginate (referred to as “alginate”) (Sigma Aldrich, CAS-number 9005-38-3) or 0.1% (weight/volume) digested alginate. At these concentrations, both alginate and digested alginate are soluble in the culture medium. The digested alginate was produced by enzymatically digesting 2% alginate with 1 unit ml⁻¹ of alginate lyase (Sigma Aldrich, CAS-number 9024-15-1) at 37 °C for 48 hours. Carbon sources were prepared in nanopure water and filter sterilized using 0.40 µm Surfactant-Free Cellulose Acetate filters (Corning, USA). Well-mixed batch experiments in alginate or digested alginate medium were performed in 96-well plates (Greiner Bio) and growth dynamics were measured using a microwell plate reader (Biotek, USA). Plate reader assays were initiated as described previously⁶⁷. Briefly, 1 ml from a culture grown for 18 h on Marine broth was centrifuged at 5000 g in 1.5 ml microfuge tubes for 5 minutes. The supernatant was discarded and the cell pellet was subjected to two rounds of washing with the basal TR salts medium. The cell-pellet was then resuspended in 1 ml of TR salts medium and 5 µl of this suspension inoculated into 195 µl TR medium with either carbon source (~10⁵ colony forming units (CFUs ml⁻¹) in a 96-well plate (Greiner Bio). The optical density (600 nm) was then measured every 15 minutes for 40 hours. All measurements had six biological replicates.

Alginate oligosaccharide measurements

Oligosaccharide measurements were performed using liquid chromatography time of flight mass spectrometry (LC-QTOF-MS). Samples were prepared by diluting 1:20 in milliQ water and 5 µL of sample was injected per measurement. Chromatographic separation was performed using an Agilent 1290 stack, using an Agilent HILIC-Z column (2.7 µm particles, 2.1 x 50 mm). Mobile phase A contained 10% acetonitrile (Fisher Scientific) and 0.1% medronic acid (Agilent), and Mobile phase B contained 90% acetonitrile and 0.1% medronic acid. The separation was performed as follows: Mobile phase B 100% for 1 minute, gradient to 30% phase B over 3 minutes, 30% phase B

for 30 seconds, and equilibration of 100% phase B for 5 minutes. The flow rate was $400 \mu\text{L min}^{-1}$ at 30°C . Samples were measured using an Agilent 6520 mass spectrometer in negative mode, in 4 GHz high-resolution mode. Data analysis was performed in Agilent Quantitative Analysis software.

Microfluidics and time-lapse microscopy

Microfluidic experiments and microscopy were performed as described previously^{15,16,68}. Cells were imaged within chambers of a PDMS (Sylgard-Dow) microfluidic chip that ranged in size from $60\text{--}120 \times 60 \times 0.56 \mu\text{m}$ ($l \times b \times h$). Within these chambers, cells can attach to the glass surface and experience the medium that diffuses through lateral flow channels. Imaging was performed using IX83 inverted microscope systems (Olympus, Japan) with automated stage controller (Marzhauser Wetzlar, Germany), shutter, and laser-based autofocus system (Olympus ZDC 2). Chambers were imaged in parallel on the same PDMS chip, and phase-contrast images of each position were taken every 8 or 10 minutes. The microscopy unit and PDMS chip were maintained at 25°C using a cellVivo microscope incubation system (Pecon GmbH).

Viscosity of the alginate and digested alginate solution

We measured the viscosity of alginate solutions using shear rheology measurements. We use a 40 mm cone-plate geometry (4° cone) in a Netzsch Kinexus Pro+ rheometer. 1200 μL of sample was placed on the bottom plate, the gap was set at 150 μm and the sample trimmed. We used a solvent trap to avoid sample evaporation during measurement. The temperature was set to 25°C using a Peltier element. We measure the dynamic viscosity over a range of shear rates = $0.1\text{--}100 \text{ s}^{-1}$. We report the viscosity of each solution as the average viscosity measured over the shear rates $10\text{--}100 \text{ s}^{-1}$, where the shear-dependence of the viscosity was low.

We measured the viscosity of 0.1% (w/V) alginate dissolved in TR media, which was $1.03 \pm 0.01 \text{ mPa}\cdot\text{s}$ (reporting the mean and standard deviation of three technical replicates.). The viscosity of 0.1% digested alginate in TR media was found to be $0.74 \pm 0.01 \text{ mPa}\cdot\text{s}$. This means that the viscosity of alginate in our microfluidic experiments is 36% higher than of digested alginate, but the viscosities are close to those expected of water ($0.89 \text{ mPa}\cdot\text{s}$ at 25°C according to Berstad and colleagues (1988)¹⁸).

Motility assays

Cells were grown for 10 hours in Marine Broth (DIFCO) after which 10 μL of culture was used to inoculate culture tubes (Greiner) containing 5 ml of TR medium with either 0.1% alginate or 0.1% digested alginate. After 6 hours of growth at 25°C , 2 μL of cell suspension was inoculated into microfluidic growth chambers. Cells within six replicate chambers were then imaged with the phase-contrast channel at a high frame rate (125 Hz, i.e., frames s^{-1}) using the same microscopy setup described above.

Chemotaxis assays

To assess whether polymeric alginate and digested alginate attract *Vibrio cyclitrophicus* ZF270, we used the *In Situ* Chemotaxis Assay^{21,22} (ISCA), a microfluidic device consisting of a 5×5 array of microwells that can be individually loaded with solutions of different chemicals (110 μL each). Once the ISCA is deployed in an aqueous environment, the chemicals diffuse out of the wells through a small port, creating chemical gradients which will guide chemotactic bacteria inside the wells of the device^{21,22}. *Vibrio cyclitrophicus* ZF210 was plated on Marine Agar (BD Difco) from a glycerol stock and grown for 16 h at 27°C . A single colony was then incubated in 10% Marine Broth (BD Difco) in 0.22- μm filtered artificial seawater (Instant Ocean, Spectrum Brands) and grown overnight at 27°C and 180 rpm. The culture was diluted down to $1 \times 10^6 \text{ cells mL}^{-1}$ in 0.22 μm filtered artificial seawater (Instant Ocean, Spectrum Brands) to perform the chemotaxis experiment. Both chemoattractants (alginate and digested alginate) were diluted in sterile seawater (35 g l^{-1} ; Instant Ocean, Spectrum Brands) at a final concentration of 0.1% and then

filtered with a 0.2 μm filter (Millipore) to remove particles and potential contaminants. Within the ISCA, one full row of five wells was used per chemoattractant as technical replicates. The chemoattractants were injected in triplicate ISCA with a sterile 1 ml syringe (Codau) and needle (27 G, Henke Sass Wolf). A last row containing 0.2 μm -filtered seawater acted as negative control accounting for cells swimming in the device by random motility only. Experiments were conducted by incubating the ISCAs for 1 h in the diluted *Vibrio cyclitrophicus* ZF270 culture. Upon time completion, a sterile syringe and needle were used to retrieve the content of the wells and transferred to 1 ml microfuge tubes resulting in a pooling of a row of 5 wells containing the same sample. Sample staining was performed with SYBR Green I (Thermo Fisher) and the chemotactic response was quantified by counting cells using flow cytometry. The strength of the chemotactic response was determined by the mean chemotactic index (I_C), defined as the ratio of the number of cells found in each chemoattractant to the number of cells in control wells containing filtered seawater (so that attraction corresponds to $I_C > 1$).

Culturing and harvesting cells for transcriptomics

Cells were grown for 18 hours in Marine Broth (DIFCO) after which 1 ml of culture was centrifuged at 5000 g in 1.5 ml microfuge tubes for 5 minutes. The supernatant was discarded and the cell pellet was subjected to two rounds of washing with the basal TR salts medium. The cell pellet was then resuspended in 1 ml of TR salts medium and 250 μl of this suspension were used to inoculate 100 ml flasks (Schott-Duran) containing 10 ml of TR medium with either 0.1% alginate or 0.1% digested alginate. This was done in parallel for six flasks. Once cultures in the flasks reached mid-exponential phase (10 hours and 15 hours after inoculation for digested alginate and alginate, respectively) and had approximately the same OD (0.39 for digested alginate and 0.41 for alginate), 2 ml of cultures were harvested for RNA extraction. Samples were stabilized with the RNprotect reagent (Qiagen) and RNA was extracted using the RNeasy mini kit (Qiagen).

Sequencing and gene annotation of *Vibrio cyclitrophicus* ZF270

Long read sequencing using the Oxford Nanopore Platform (Long read DNA sequencing kit) and short read sequencing using the Illumina platform (Illumina DNA Prep kit and IDT 10bp UDI indices, and sequenced on an Illumina NextSeq 2000 producing 2x151bp reads) was performed by the Microbial Genome Sequencing Center, Pittsburgh, USA (MiGS), to create a new closed reference genome of *V. cyclitrophicus* ZF270 (BioProject PRJNA991487). Annotation of this genome was done with RASTtk (v2.0, Rapid Annotation using Subsystem Technology tool kit⁶⁹). Additionally, KEGG Ontology identifiers ("K numbers") were annotated with BlastKOALA (v2.2)⁷⁰. Dedicated annotation of alginate lyase genes was performed by homology search for proteins belonging to the PL5, PL6, PL7, PL14, PL15, PL17, PL18, PL31, PL36, or PL39 family⁷¹ by dbCAN2 (v9.0)⁷². Enzymes for alginate transport and metabolism were identified by BLASTn-search^{73,74} of gene sequences of *Vibrio splendidus* 12B01, which were previously identified as minimum genetic prerequisites for alginate utilization and enabled alginate degradation when cloned into *E. coli*³¹.

Location prediction of alginate lyases

Signal peptides were annotated using SignalP (v.5.0)⁷⁵, and LipoP (v.1.0)⁷⁶. SignalP discriminated between 1) Sec/SPI: "standard" secretory signal peptides transported by the Sec translocon and cleaved by Signal Peptidase I (SPI), 2) Sec/SPII: lipoprotein signal peptides transported by the Sec translocon and cleaved by Signal Peptidase II (SPII), and 3) Tat/SPI: Tat signal peptides transported by the Tat translocon and cleaved by SPI. LipoP discriminates between 1) SPI: signal peptide, 2) SPII: lipoprotein signal peptide, and 3) TMH: n-terminal transmembrane helix. All predictions were in agreement, apart from one PL7 (gene 1136176.5.peg.4375) which was predicted by LipoP as cytoplasmic and by SignalP as equally likely cytoplasmic as containing a lipoprotein signal peptide.

Transcriptomic analysis: Sequencing, pre-processing, differential expression analysis, and functional analysis

Sequencing (12 M reads, 2 x 50 bp) of the isolated RNA was performed by MiGS after rRNA depletion using RiboZero Plus (Illumina). cDNA libraries were prepared using an Illumina DNA Prep kit and IDT 10 bp UDI indices, and sequenced on an Illumina NextSeq 2000. Preprocessing of the raw reads was carried out as follows: Quality control was performed with FastQC (v0.11.9)⁷⁷ and reads were trimmed with Trimmomatic (v0.38)⁷⁸; the high-quality reads were mapped to the reference genome (described above) with Bowtie2 (v2.3.5.1)⁷⁹; binarization, sorting, and indexing were done with Samtools (v1.10)⁸⁰; gene counts were computed with the featureCount function of Subread (v2.0.1)⁸¹.

Differential expression analysis was performed with DESeq2 (v1.30.1)⁸². In brief, DESeq2 normalizes the raw read counts with normalization factors ("size factors") to account for differences in sequencing depth between samples. Subsequently, gene-wise dispersion estimates are computed for each gene separately using maximum likelihood, and then shrunk toward the values predicted by the dispersion-mean dependence curve to obtain final dispersion values. Finally, DESeq2 fits a negative binomial model to the read counts and performs significance testing using the Wald test. Here reported *p*-values result from the Wald test of read counts from the digested alginate condition compared to the alginate condition and were adjusted for multiple testing by Benjamini-Hochberg correction⁸³ as implemented in the `p.adjust` function of base R (v4.1.2). The reported log2 fold changes indicate the log2(DESeq2-normalised reads in digested alginate condition / DESeq2-normalised reads in alginate condition) for each gene.

Visualization of gene maps was performed in R with the `ggplot2` package (v3.4.0) and the extension `gggenes` (v0.4.1) by David Wilkins.

For systematic functional analysis we performed gene set enrichment analysis (GSEA)²³ using the `fgsea` function of the `fgsea` package (v1.20.0) with minimal number of unique genes per gene set "minSize" = 5 and number of permutations "nPermSimple" = 1000000. In brief, GSEA takes the full gene list ranked by log2 fold change and annotated with K numbers as input and determines whether the member genes of any KEGG pathway are randomly distributed throughout the ranked gene list or whether they are primarily found at the top or bottom²³. This is quantified by the enrichment score (ES), which corresponds to a weighted Kolmogorov-Smirnov-like statistic. The ES of each gene set is normalized to the mean enrichment of random samples of the same size to account for the size of the set, yielding the normalized enrichment score (NES). To estimate the significance level of the enrichment score, the *p*-value of the observed enrichment score is calculated relative to a null distribution that was computed from permuted data. The estimated significance level was adjusted to account for multiple hypothesis testing. As gene sets we chose all KEGG pathways and KEGG BRITE categories (as noted in Table S1 in column "KEGG_pathway") within all genes of *V. cyclitrophicus* ZF270 annotated with a KEGG Ontology identifier ("K number"). Visualization of differential expression levels in KEGG pathways was performed with the R package `pathview` (v1.35.0)⁸⁴. We also formed gene sets of the genes associated with alginate utilization (see "Sequencing and gene annotation of *Vibrio cyclitrophicus* ZF270", Table S16), of the genes of the flagellar locus that map to the KEGG pathways "Bacterial motility proteins", and of the genes of the flagellar locus that map to the KEGG pathways "Bacterial chemotaxis" (as noted in Table S17) within all genes of *V. cyclitrophicus* ZF270.

Image analysis

Cells within microscopy images were segmented and tracked using `ilastik` (v1.3) ("pixel classification workflow" and "tracking with probabilities workflow"). Phase contrast images were used for alignment, segmentation, tracking and linking. Images were cropped at the boundaries of each microfluidic chamber. The lineage identity of each single cell was assigned by `ilastik`'s

tracking plugin and visualized by coloring the segmented cells, respectively. The growth rate of each cell was computed as the change of cell area over time, i.e., via a linear regression of the single-cell area over the time between consecutive cell divisions, based on ilastik's segmentation and tracking output. Cells that were tracked over less than three frames were excluded. Measurement of swimming speeds and displacement of cells was performed using ilastik (v1.4), ImageJ (v2.3) and Trackmate (v7.5.2). Briefly, cell-segmentation ("pixel classification workflow" in ilastik) and tracking ("animal tracking workflow" in ilastik) were performed using the high frame rate phase contrast images in ilastik (v1.4). Cell trajectories and properties were then computed using the output of the ilastik workflow in Trackmate.

Dispersal analysis

For the analysis on the dispersal of cells (**Fig. 2** [↗](#)), we computed the cell number as the total number of cells within a microfluidic chamber. The change in the number of cells was computed by subtracting the number of cells before the medium switch (i.e., average number of cells between $t = 1.9$ to 2.1 h) from the number of cells after the medium switch (i.e., average number of cells between $t = 3.9$ to 5.5 h).

Datasets and statistical analysis

All batch experiments were replicated 3 to 6 times. Growth curves were analyzed in Python (v3.7) using the *Amiga* package (v1.1.0)⁸⁵ and GraphPad Prism (v8, GraphPad Software). The microscopy dataset consisted of eight chambers each, corresponding to the eight replicates shown in **Figure 1** [↗](#) and **Figure 2** [↗](#). These were grouped into two biological replicates wherein each biological replicate was fed by media through a unique channel in a microfluidic chip. Cells with negative growth rates were excluded from the analysis after visual curation, as they represented artifacts, mistakes in segmentation or linking during the tracking process, or non-growing deformed cells. Each chamber was treated as an independent replicate. Comparisons were considered statistically significant when $p < 0.05$ or when the False Discovery Rate (FDR)-corrected q was smaller than 0.05. FDR corrections were applied when multiple t tests were performed for the same dataset. Measures of effect size are represented by the R^2 or η^2 value. All statistical analyses were performed in GraphPad Prism v9.0 (GraphPad Software, USA), R v4.1.2, RStudio v1.1.463 (Posit, USA).

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Data Availability

Sequencing data is available on NCBI, BioProject PRJNA991487, upon publication. All further data and code is deposited on ERIC Open (<https://opendata.eawag.ch/↗>, <https://doi.org/10.25678/0008MH↗>), available upon publication. Code will also be made publicly accessible on <https://github.com/Microbial-Systems-Ecology/publications↗> upon publication.

Competing interests

The authors declare no competing interests.

Author Contributions

GD, AS, JS, and JK conceived the research along with MA. GD designed and performed the experiments. AS developed and implemented the algorithms to compute single-cell growth rates and visualize lineage identities from segmented and tracked cells. AS analyzed and visualized the RNA-seq data. JK conceived and implemented the cell dispersal analysis. EC conducted the ISCA experiment. SP carried out the mass spectrometry measurements. AS and GD analyzed the data with inputs from JK, JS, CM, MA, and RS. AS wrote the manuscript with inputs from GD, OS, MA, JK, JS, EC, SP, CM, and RS.

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Editors

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Reviewer #1 (Public Review):

Summary:

The authors attempt to understand how cells forage for spatially heterogeneous complex polysaccharides. They aimed to quantify the foraging behavior and interrogate its genetic basis. The results show that cells aggregate near complex polysaccharides and disperse when simpler byproducts are added. Dispersing cells tend to move towards the polysaccharide. The authors also use transcriptomics to attempt to understand which genes support each of these

behaviors - with motility and transporter related genes being highly expressed during dispersal, as expected.

Strengths:

The paper is well written and builds on previous studies by some of the authors showing similar behavior by a different species of bacteria (*Caulobacter*) on another polysaccharide (xylan). The conceptual model presented at the end encapsulates the findings and provides an interesting hypothesis. I also find the observation of chemotaxis towards the polysaccharide in the experimental conditions interesting.

Weaknesses:

Much of the genetic analysis, as it stands, is quite speculative and descriptive. I found myself confused about many of the genes (e.g., quorum sensing) that pop up enriched during dispersal quite in contrast to my expectations. While the authors do discuss this in the text as worth following up on, I think the analysis as it stands is speculative about the behaviors observed. In the authors' defense, I acknowledge that it might have the potential to generate hypotheses and thus aid future studies.

<https://doi.org/10.7554/eLife.93855.2.sa3>

Reviewer #2 (Public Review):

Summary:

The paper sets out to understand the mechanisms underlying the colonization and degradation of marine particles using a natural *Vibrio* isolate as a model. The data are measurements of motility and gene expressing using microfluidic devices and RNA sequencing. The results reveal that degradation products of alginate do stimulate motility but not chemotaxis. In contrast, alginate itself (the polymer) does stimulate chemotaxis. Further, the dispersal from degrading alginate is density dependent, increasing at higher density. The evidence for these claims are strong. From these the authors propose a narrative (Fig. 6) for growth and dispersal cycles in this system. The idea is that cells colonize and degrade alginate, this degradation stimulates motility and dispersal followed by chemotaxis to a new alginate source. This complete narrative has modest support in the data. A quantitative description of these dynamics awaits future studies.

Strengths:

The microfluidic measurements are the central strength of the paper. The density dependence claim is qualitatively supported by the data. The motility and chemotaxis claims are also well supported by the data. The presentation of the experiment and results are well done. The study serves to motivate a unifying picture of growth and dispersal in marine systems. This is a key process in the global carbon cycle.

Weaknesses:

Perhaps not a weakness, but a glimmer that this is not yet the full story. The RNA expression data show alginate lyase expression in response to digested alginate which is unexpected given the narrative articulated above. Why express lyases while leaving the polymer patch via motility? This question is addressed in the Discussion. A holistic and quantitative picture of the proposed process in Figure 6 awaits additional studies.

<https://doi.org/10.7554/eLife.93855.2.sa2>

Reviewer #3 (Public Review):**Summary:**

In this manuscript, Stubbusch and coauthors examine the foraging behavior of a marine species consuming an abundant marine polysaccharide. Laboratory experiments in a microfluidic setup are complemented with transcriptomic analyses aiming at assessing the genetic bases of the observed behavior. Bacterial cells consuming the polysaccharide form cohesive aggregates, while start dispersing away when the byproduct of the digestion of the polysaccharide start accumulating. Dispersing cells, tend to be attracted by the polysaccharide. Expression data show that motility genes are enriched during the dispersal phase, as expected. Counterintuitively, in the same phase, genes for transporters and digestions of polysaccharide are also highly expressed.

Strengths:

The manuscript is very well written and easy to follow. The topic is interesting and timely. The genetic analyses provide a new, albeit complex, angle to the study of foraging behaviors in bacteria, adding to previous studies conducted on other species.

Weaknesses:

I find this paper very descriptive and speculative. The results of the genetic analyses are quite counterintuitive; therefore, I understand the difficulty of connecting them to the observations coming from experiments in the microfluidic device. However, they could be better placed in the literature of foraging - dispersal cycles, beyond bacteria. In addition, the interpretation of the results is sometimes confusing.

<https://doi.org/10.7554/eLife.93855.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Editors' recommendations for the authors

The reviewers recommend the following:

- (a) Digging deeper into the discussion of the density-dependent dispersal.*
- (b) Clarifying the microfluidic setup.*
- (c) Clarifying the description and interpretation of the transcriptomic evidence.*
- (d) Toning down carbon cycle connections (some reviewers felt the evidence did not fully support the claims).*

We would like to thank the editors for their thoughtful evaluation of our manuscript and their clear suggestions. We have revised the manuscript in the light of these comments, as we outline below and address in detail in the point-by-point response to the reviewers' comments that follows.

(a) We have expanded the discussion of density-dependent dispersal and revised Figure 2C to improve clarity.

(b) We have also added further information concerning the microfluidic setup in the results section and provide an illustration of the setup in a new figure panel, Figure 1A.

(c) Addressing the reviewers' comments on the transcriptomic analysis, we have added more information in the description and interpretation of the results.

(d) We have rephrased the text describing the role of degradation-dispersal cycles for carbon cycling to highlight it as the motivation of this study and emphasize the link to literature on foraging, without creating expectations of direct measurements of global carbon cycling.

Public Reviews:

Reviewer #1 (Public Review):

[...]

Weaknesses:

Much of the genetic analysis, as it stands, is quite speculative and descriptive. I found myself confused about many of the genes (e.g., quorum sensing) that pop up enriched during dispersal quite in contrast to my expectations. While the authors do mention some of this in the text as worth following up on, I think the analysis as it stands adds little insight into the behaviors studied. However, I acknowledge that it might have the potential to generate hypotheses and thus aid future studies. Further, I found the connections to the carbon cycle and marine environments in the abstract weak --- the microfluidics setup by the authors is nice, but it provides limited insight into naturalistic environments where the spatial distribution and dimensionality of resources are expected to be qualitatively different.

We thank the reviewer for their suggestions to improve our manuscript. We agree that the original manuscript would have benefitted from more detailed interpretation of the observed changes in gene expression. We have revised the manuscript to elaborate on the interpretation of the changes in expression of quorum sensing genes (see response to reviewer 1, comment 3), motility genes (see response to reviewer 1, comment 6), alginate lyase genes (see response to reviewer 1, comment 7 and reviewer 2, comment 2), and ribosomal and transporter genes (see response to reviewer 2, comment 2).

In general, we think that the gene expression study not only supports the phenotypic observations that we made in the microfluidic device, such as the increased swimming motility when exposed to digested alginate medium, but also adds further insights. Our reasoning for studying the transcriptomes in well mixed-batch cultures was the inability to study gene expression dynamics to support the phenotypic observations about differential motility and chemotaxis in our microfluidics setup. The transcriptomic data clearly show that even in well-mixed environments, growth on digested alginate instead of alginate is sufficient to increase the expression of motility and chemotaxis genes. In addition, the finding that expression of alginate lyases and metabolic genes is increased during growth on digested alginate was revealed through the analysis of transcriptomes, something which would not have been possible in the microfluidic setup. We agree with the reviewer that our analyses implicate further, perhaps unexpected, mechanisms like quorum sensing in the cellular response to breakdown products, and that this represents an interesting avenue for further studies.

Finally, we also agree with the reviewer that it would be good to be more explicit in the text that our microfluidic system cannot fully capture the complex dynamics of natural environments. Our approach does, however, allow the characterization of cellular behaviors at spatial and temporal scales that are relevant to the interactions of bacteria, and thus provides a better understanding of colonization and dispersal of marine bacteria in a manner that is not possible through in situ experiments. We have edited our manuscript to highlight this and modified our statements regarding carbon cycling towards emphasizing the role

degradation-dispersal cycles in remineralization of polysaccharides (see response to reviewer 1, comment 2).

Reviewer #2 (Public Review):

[...]

Weaknesses:

The explanation of the microfluidics measurements is somewhat confusing but I think this could be easily remedied. The quantitative interpretation of the dispersal data could also be improved and I'm not clear if the data support the claim made.

We thank the reviewer for their comments and helpful suggestions. We have revised the manuscript with these suggestions in mind and believe that the manuscript is improved by a more detailed explanation of the microfluidic setup. We have added more information in the text (detailed in response to reviewer 2, comments 1 and 2) and have added a depiction of the microfluidic setup (Fig. 1A). We have also modified the presentation and discussion of the dispersal data (Fig. 2C), as described in detail below in response to reviewer 2, comment 4, and argue that they clearly show density-dependent dispersal. We believe that this modification of how the results are presented provides a more convincing case for our main conclusion, namely that the presence of degradation products controls bacterial dispersal in a density-dependent manner.

Reviewer #3 (Public Review):

[...]

Weaknesses:

I find this paper very descriptive and speculative. The results of the genetic analyses are quite counterintuitive; therefore, I understand the difficulty of connecting them to the observations coming from experiments in the microfluidic device. However, they could be better placed in the literature of foraging - dispersal cycles, beyond bacteria. In addition, the interpretation of the results is sometimes confusing.

We thank the reviewer for their suggestions to improve the manuscript. We have edited the manuscript to interpret the results of this study more clearly, in particular with regard to the fact that breakdown products of alginate cause cell dispersal (see response to reviewer 2, comment 1), gene expression changes of ribosomal proteins and transporters (see response to reviewer 2, comment 2), as well as genes relating to alginate catabolism (see response to reviewer 2, comment 3).

To provide more context for the interpretation of our results we now also embed our findings in more detail in the previous work on foraging strategies and dispersal tradeoffs.

Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

(1) The authors should clarify in more detail what they mean by density dependence in Figure 2. Usually density dependence refers to a per capita dependence, but here it seems that the per capita rate of dispersal might be roughly independent of density (Figure 2c; if you double the number of cells it doubles the number of cells leaving). Rather it seems the dispersal is such that the density of remaining cells falls below a threshold (~300 cells).

We thank the reviewer for raising this important point. To analyze the data more explicitly in terms of per capita dependence and so make the density dependence in the dispersal from the microfluidic chambers more clear, we have modified Figure 2C and edited the text.

In the modified Figure 2C, we computed the fraction of dispersed cells for each chamber (i.e. the change in cell number divided by the cell number at the time of the nutrient switch). This quantity directly reveals the per-capita dependence, as mentioned by reviewer 1, and is now represented on the y-axis of Figure 2C instead of the absolute change in cell number.

These data demonstrate that the fraction of dispersed cells increases with increasing numbers of cells present in the chamber at the time of switching, with more highly populated chambers showing a higher fraction of dispersed cells. These findings indicate that there is a strong density dependence in the dispersal process.

As pointed out by reviewer 1, another interesting aspect of the data is the transition at low cell number. The fraction of dispersed cells is negative in the case of the chamber with approximately 70 cells, consistent with no dispersal at this low density, and a moderate density increase as a function of continued growth.

In addition to the new analysis presented in Figure 2C, we have modified the paragraph that discusses this result as follows (line 208):

“We indeed found that the nutrient switch caused a few or no cells to disperse from small cell groups (Fig. 2B), whereas a large fraction of cells from large cell groups dispersed (Fig. 2C). In fact, the fraction of cells that dispersed upon imposition of the nutrient switch showed a strong positive relationship with the number of cells present, meaning that cells in chambers with many cells were more likely to disperse than cells in chambers with fewer cells (Fig. 2C).”

(2) The authors should tone down their claims about the carbon cycle in the abstract. I do not believe the results as they stand could be used to understand degradation-dispersal cycles in marine environments relevant to the carbon cycle, since these behaviors have been studied in microfluidic environments which in my understanding are quite different. As such, statements such as "degradation-dispersal cycles are an integral part in the global carbon cycle, we know little about how cells alternate between degradation and motility" and "Overall, our findings reveal the cellular mechanisms underlying bacterial degradation-dispersal cycles that drive remineralization in natural environments" are overstated in the abstract.

We appreciate the reviewer’s comments regarding the connections of our work with the carbon cycle. We have now rephrased these statements in our manuscript to describe a potential connection between our work and the marine carbon cycle. The colonization of polysaccharides particles by bacteria and subsequent degradation has been widely acknowledged to play a significant role in controlling the carbon flow in marine ecosystems. (Fenchel, 2002; Preheim et al., 2011; Yawata et al., 2014, 2020). We still refer to carbon flow in the revised manuscript, though cautiously, as microbial remineralization of biomass, which is recognized as an important factor in the marine biological carbon pump (e.g., (Chisholm, 2000; Jiao et al., 2024). As stated in the previous version of the manuscript, the main motivation of our work was to study the growth behaviors of marine heterotrophic bacteria during polysaccharide degradation, especially to understand when bacteria depart already colonized and degraded particles and find novel patches to grow and degrade, a process that is poorly understood. Therefore, it is conceivable that degradation-dispersal cycles do play a role in the flow of carbon in marine ecosystems. However, we acknowledge that the carbon cycle is influenced by a multitude of biological and chemical processes, and the bacterial degradation-dispersal cycle might not be the sole mechanism at play.

We also appreciate the reviewer's comments highlighting that the complexity of natural environments is not fully captured in our microfluidics system. However, our microfluidics setup does allow us to quantify responses and behaviors of microbial groups at high spatial and temporal resolution, especially in the context of environmental fluctuations. Microbes in nature interact at small spatial scales and have to respond to changes in the environment, and the microfluidics setup enables the quantification of these responses. Moreover, dispersal of the bacterium *V. cyclitrophicus* that we use in our study, has been previously observed even during growth on particulate alginate (Alcolombri et al., 2021), but the cues and regulation controlling dispersal behaviors have been unclear. Microfluidic experiments have now allowed us to study this process in a highly quantitative manner, and align well with observations from experiments from more nature-like settings. These quantitative experiments on bacterial strains isolated from marine particles are expected to constrain quantitative models of carbon degradation in the ocean (Nguyen et al., 2022).

We have now adjusted our statements throughout our manuscript to reflect the knowledge gaps in understanding the triggers of degradation-dispersal cycles and their links with carbon flow in marine ecosystems. The revised manuscript, especially, contains the following statements (line 47 and line 60):

"Even though many studies indicate that these degradation-dispersal cycles contribute to the carbon flow in marine systems, we know little about how cells alternate between polysaccharide degradation and motility, and which environmental factors trigger this behavioral switch."

"Overall, our findings reveal cellular mechanisms that might also underlie bacterial degradation-dispersal cycles, which influence the remineralization of biomass in marine environments."

(3) The authors should clarify why they think quorum-sensing genes are increased in expression on digested alginate. The authors currently mention that QS could be used to trigger dispersal, but given the timescales of dispersal in Figure 2 (~half an hour), I find it hard to believe that these genes are expressed and have the suggested effect on those timescales. As such I would have expected the other way round - for QS genes to be expressed highly during alginate growth, so that density could be sensed and responded to. Please clarify.

We have now clarified this point in the revised manuscript. While the triggering of dispersal by quorum-sensing genes may indeed appear counterintuitive, and the response is rapid (we see dispersal of cells within 30-40 minutes), both observations are in line with previous studies in another model organism *Vibrio cholerae*. The dispersal time is similar to the dispersal time of *V. cholerae* cells from biofilms, as described by Singh and colleagues, (Figure 1E of Ref. Singh et al., 2017). In that case, induction of the quorum sensing dispersal regulator HapR was observed during biofilm dispersal within one hour after switch of condition (Fig. 2, middle panel of Ref. Singh et al., 2017). Even though the specific quorum sensing signaling molecules are probably different in our strain (there is no annotated homolog of the hapR gene in *V. cyclitrophicus*), we observed that the full set of quorum sensing genes was enriched in cells growing on digested alginate (as reported in line 314 and Fig. 4A).

We have added this information in the manuscript (line 317):

"The set of quorum sensing genes was also positively enriched in cells growing on digested alginate (Fig. 4A and S4F, Table S13). This role in dispersal is in agreement with a previous study that showed induction of the quorum sensing master regulator in *V. cholerae* cells during dispersal from biofilms on a similar time scale as here (less than an hour) [28]."

Reviewer #2 (Recommendations For The Authors):

(1) Around line 144 - I don't really understand how you flow alginate through the microfluidic platform. It seems if the particles are transiently going through the microfluidic chamber then the flow rate and hence residence time of the alginate particles will matter a lot by controlling the time the cells have to colonize and excrete enzymes for alginate breakdown. Or perhaps the alginate is not particulate but is instead a large but soluble polymer? I think maybe a schematic of the microfluidic device would help -- there is an implicit assumption that we are familiar with the Dal Co et al device, but I don't recall its details and maybe a graphic added to Figure 1 would help.

a. In reviewing the Dal Co paper I see that cells are trapped and the medium flows through channels and the plane where the cells are held. I am still a little confused about the size of the polymeric alginate -- large scale (>1µm) particles or very small polymers?

We have now provided a detailed description of our microfluidic experimental system. At the start of the experiments, cells are in fact not trapped within the microfluidic device, but grow and can move freely within a chamber designed with dimensions (sub-micron heights) so that growth occurs only as a monolayer. Cells were exposed to nutrients, either alginate or alginate digestion products, both in soluble form (not particles). These compounds were flowed into the device through a main channel, but entered the flowfree growth chambers by diffusion. To make these aspects of our experiments clearer, we have added further information on this in the Materials & Methods section (line 556), added this information in the abstract (line 51), and in the results (line123).

To make our microfluidic setup clearer, we have followed this advice and added a schematic as Figure 1A and have added more information on the setup to the main text (line 153):

“In brief, the microfluidic chips are made of an inert polymer (polydimethylsiloxane) bound to a glass coverslip. The PDMS layer contains flow channels through which the culture medium is pumped continuously. Each channel is connected to several growth chambers that are laterally positioned. The dimensions of these growth chambers (height: 0.85 µm, length: 60 µm, width: 90-120 µm) allow cells to freely move and grow as monolayers. The culture medium, containing either alginate or digested alginate in their soluble form, is constantly pumped through the flow channel and enters the growth chambers primarily through diffusion [15,16,4,17,8]. Therefore, the number of cells and their positioning within microfluidic chambers is determined by the cellular growth rate as well as by cell movement⁴. This setup combined with time-lapse microscopy allowed us to follow the development of cell communities over time.”

(2) What makes this confusing is the difference between Figure 1C and Figure S2A -- the authors state that the difference in Figure 1C is due to dispersal, but is there flow through the microfluidic device? So what role does that flow through the device have in dispersal? Is the adhesion of the cell groups driven at all by a physical interaction with high molecular weight polymers in the microfluidic devices or is this purely a biological effect? Could this also be explained by different real concentrations of nutrients in the two cases?

We realize from this comment that the role of flow of the medium in the microfluidic setup was not clearly addressed in our manuscript. In fact, cells were not exposed to flow, and nutrients were provided to the growth chambers by diffusion. We have added a clearer explanation of this point on line 158:

“The culture medium, containing either alginate or digested alginate in their soluble form, is constantly pumped through the flow channel and enters the growth chambers primarily

through diffusion [15,16,4,17,8]. Therefore, the number of cells and their positioning within microfluidic chambers is determined by the cellular growth rate as well as by cell movement4.“

One purely physical effect that we anticipate is that a high viscosity of the medium could immobilize cells. To address this point, we measured the viscosity of both alginate and digested alginate and conclude that the increase in viscosity is not strong enough to immobilize cells. We added a statement in the text (line 170)

“To test the role of increased viscosity of polymeric alginate in causing the increased aggregation of cells, we measured the viscosity of 0.1% (w/v) alginate or digested alginate dissolved in TR media. For alginate, the viscosity was 1.03 ± 0.01 mPa·s (mean and standard deviation of three technical replicates) whereas the viscosity of digested alginate in TR media was found to be 0.74 ± 0.01 mPa·s. Both these values are relatively close to the viscosity of water at this temperature (0.89 mPa·s [18]) and, while they may affect swimming behavior [19], they are insufficient to physically restrain cell movement [20].”

as well as a section in the Materials and Methods (line 594):

“Viscosity of the alginate and digested alginate solution

We measured the viscosity of alginate solutions using shear rheology measurements. We use a 40 mm cone-plate geometry (4° cone) in a Netzsch Kinexus Pro+ rheometer. 1200 μ L of sample was placed on the bottom plate, the gap was set at 150 μ m and the sample trimmed. We used a solvent trap to avoid sample evaporation during measurement. The temperature was set to 25°C using a Peltier element. We measure the dynamic viscosity over a range of shear rates = 0.1 – 100 s⁻¹. We report the viscosity of each solution as the average viscosity measured over the shear rates 10 – 100 s⁻¹, where the shear-dependence of the viscosity was low.

We measured the viscosity of 0.1% (w/V) alginate dissolved in TR media, which was 1.03 ± 0.01 mPa·s (reporting the mean and standard deviation of three technical replicates.). The viscosity of 0.1% digested alginate in TR media was found to be 0.74 ± 0.01 mPa·s. This means that the viscosity of alginate in our microfluidic experiments is 36% higher than of digested alginate, but the viscosities are close to those expected of water (0.89 mPa·s at 25 degree Celsius according to Berstad and colleagues [18]).”

While our microfluidic setup allows us to track the position and movement of cells in a spatially structured setting, these observations do not allow us to distinguish directly whether the differences in dispersal are a result of purely physical effects of polymers on cells or are a result of them triggering a biological response in cells that causes them to become sessile. It is known that bacterial appendages like pili interact with polysaccharide residues (Li et al., 2003). Therefore, it is quite plausible that cross-linking by polysaccharides can contribute growth behaviors on alginate. However, our analysis of gene expression demonstrates that flagellum-driven motility is decreased in the presence of alginate compared to digested alginate, alongside other major changes in gene expression. In addition, our measures of dispersal show that dispersal of cells when exposed to digested alginate is density dependent. Both observations suggest that the patterns in dispersal are governed by decision-making processes by cells resulting in changes in cell motility, rather than being a product of purely physical interactions with the polymer.

The finding that viscosities of both alginate and digested alginate are similar to that of water, suggests that diffusion of nutrients in the growth chambers should be similar. Therefore, we think that the differences in real concentrations of nutrients is likely not contributing to the observed differences in behavior.

(3) Why is Figure S1 arbitrary units? Does this have to do with the calibration of LC-MS? It would be better, it seems, to know the concentrations in real units of the monomer at least.

We agree with the reviewer that it would have been better to have absolute concentrations for these compounds. However, to calibrate the mass spectrometer signals (ion counts) to absolute concentrations for the different alginate compounds, we would need an analytical standard of known concentration. We are not aware of such a standard and thus report only relative concentrations. We agree that the y-axis label of Figure S1 should not contain ‘arbitrary’ units, as it shows a ratio (of measurements in the same arbitrary units). We have edited the labels of Figure S1 accordingly and the figure legend in line 26 of the Supplemental Material (“Relative concentrations...”).

(4) Line 188 - density-dependent dispersal. The claim here is that "cells in chambers with many cells were more likely to disperse than cells in chambers with less cells." (my emphasis). Looking at the data in Figure 2C it appears that about 40% of the cells disperse irrespective of the density, before the switch to digested alginate. So it would seem that there is not a higher likelihood of dispersal at higher cell densities. For the very highest cell density, it does appear that this fraction is larger, but I'd be concerned about making this claim from what I understand to be a single experiment. To support the claim made should the authors plot Change in Cell number/Starting Cell number on the y-axis of Fig. 2C to show that the fraction is increasing? It would seem some additional data at higher starting cell densities would help support this claim more strongly.

We thank the reviewer for this comment, which is in line with a remark made by reviewer 1 in their comment 1. In response to these two comments (and as described above), we have edited Figure 2C and now have plotted the change in cell number relative to starting cell number at the y axis to directly show the density dependence. We observe a positive (approximately linear) relationship between the fraction of dispersed cells with the number of cells present in the chamber at the time of switching. This indicates that there is a density dependence in the dispersal process, with highly populated chambers showing a higher fraction of dispersed cells.

In addition to the change in Figure 2C, we have modified the paragraph around line 208: “We indeed found that the nutrient switch caused a few or no cells to disperse from small cell groups (Fig. 2B), whereas a large fraction of cells from large cell groups dispersed (Fig. 2C). In fact, the fraction of cells that dispersed upon imposition of the nutrient switch showed a strong positive relationship with the number of cells present, meaning that cells in chambers with many cells were more likely to disperse than cells in chambers with fewer cells (Fig. 2C).”

The highest cell number at the start of the switch that we include is about 800 cells. The maximum number of cells that can fit into a chamber are ca. 1000 cells. Thus, 800 resident cells are close to the maximal density.

(5) A comment -- I find the result of significant chemotaxis towards alginate but not the monomers of alginate to be quite surprising. The ecological relevance of this (line 219) seems like an important result that is worth expanding on a bit at least in the discussion. For now, my question is whether the authors know of any mechanism by which chemotaxis receptors could respond to alginate but not the monomer. How can a receptor distinguish between the two?

We agree that this result is surprising, given that oligomers can be more easily transported into the periplasm where sensing takes place, and they also provide an easier accessible

nutrient source. Indeed, in case of the insoluble polymer chitin it has been shown that chemotaxis towards chitin is mediated by chitin oligomers (Bassler et al., 1991), which was suggested as a general motif to locate polysaccharide nutrient sources (Keestra et al., 2022). However, a recent study has changed this perspective by showing widespread chemotaxis of marine bacteria towards the glucose-based marine polysaccharide laminarin, but not towards laminarin oligomers or glucose (Clerc et al., 2023). Together with our results on chemotaxis towards alginate (but not significantly toward alginate oligomers) this suggests that chemotaxis towards soluble polysaccharides can be mediated by direct sensing of the polysaccharide molecules.

As recommended, we expanded the discussion of the ecological relevance and also added more information on possible mechanisms of selective sensing of alginate and its breakdown products (around line 479):

“Direct chemotaxis towards polysaccharides may facilitate the search for new polysaccharide sources after dispersal. We found that the presence of degradation products not only induces cell dispersal but also increases the expression of chemotaxis genes. Interestingly, we found that *V. cyclitrophicus* ZF270 cells show chemotaxis towards polymeric alginate but not digested alginate. This contrasts with previous findings for bacterial strains degrading the insoluble marine polysaccharide chitin, where chemotaxis was strongest towards chitin oligomers⁵³, suggesting that oligomers may act as an environmental cue for polysaccharide nutrient sources⁵⁵. However, recent work has shown that certain marine bacteria are attracted to the marine polysaccharide laminarin, and not laminarin oligomers⁵⁶. Together with our results, this indicates that chemotaxis towards soluble polysaccharides may be mediated by the polysaccharide molecules themselves. The mechanism of this behavior is yet to be identified, but could be mediated by polysaccharide-binding proteins as have been found in *Sphingomonas* sp. A1 facilitating chemotaxis towards pectin⁵⁷. Direct polysaccharide sensing adds complexity to chemosensing as polysaccharides cannot freely diffuse into the periplasm, which can lead to a trade-off between chemosensing and uptake⁵⁸. Furthermore, most polysaccharides are not immediately metabolically accessible as they require degradation. But direct polysaccharide sensing can also provide certain benefits compared to using oligomers as sensory cues. First, it could enable bacterial strains to preferably navigate to polysaccharide nutrients sources that are relatively uncolonized and hence show little degradation activity. Second, strong chemotaxis towards degradation products could hinder a timely dispersal process as the dispersal then requires cells to travel against a strong attractant gradient formed by the degradation products. Overall, this strategy allows cells to alternate between degradation and dispersal to acquire carbon and energy in a heterogeneous world with nutrient hotspots [44,59–61].”

(6) Comment on lines 287-8 -- that the "positive enrichment of the gene set containing bacterial motility proteins matched the increase in motile cells that we observe in Fig 3E." I'm confused about what is meant by the word "matched" here. Is the implication that there is some quantitative correspondence between increased motility in Figure 3 and the change in expression in Figure 4? Or is the statement a qualitative one -- that motility genes are upregulated in the presence of digested alginate? Table S12 didn't help me answer this question.

We thank the reviewer for their helpful comment. Our original statement was a qualitative one - observing that gene expression enrichment in genes associated with bacterial motility aligned with our expectations based on the previous observation of an increase in motile cells. We have now changed the wording to highlight the qualitative nature of this statement (line 315):

“The positive enrichment of the gene set containing bacterial motility proteins aligned with our expectations based on the increase in motile cells that we observed in Figure 3E (Fig. 4A,

Table S12).”

(7) Line 326 - what is the explanation for the production of public enzymes in the presence of digest? How does this square with the previous narrative about cells growing on alginate digest expressing motility genes and chemotaxing towards alginate? It seems like the story is a bit tenuous here in the sense that digested alginates stimulate both motility - which is hypothesized to drive the discovery of new alginate particles - and lyase enzymes which are used to degrade alginate. So do the high motility cells that are chemotaxing towards alginate also express lyases en route? I'm of the opinion that constructing narratives like these in the absence of a more quantitative understanding of the colonization and degradation dynamics of alginate particles presents a major challenge and may be asking more of the data than the data can provide.

a. I noted later that this is addressed later around lines 393 in the Discussion section.

Indeed, the notion that the presence of breakdown products triggers motility and also increases the expression of alginate lyases and other metabolic genes for alginate catabolism seems counterintuitive. We have now expanded our discussion of these results to contextualize these findings (around line 443):

"One reason for this observation may be that cells primarily rely on intracellular monosaccharide levels to trigger the upregulation of genes associated with polysaccharide degradation and catabolism, as has previously been observed for *E. coli* across various carbon sources [50,51]. In fact, the majority of carbon sources are sensed by prokaryotes through one-component sensors inside the cell [50]. In the one-component internal sensing scheme, the enzymes and transporters for the use of various carbon sources are expressed at basal levels, which leads to an increase in pathway intermediates upon nutrient availability. The pathway intermediates are sensed by an internal sensor, usually a transcription factor, and lead to the upregulation of transporter and enzyme expression [50,51]. This results in a positive feedback loop, which enables small changes in substrate abundance to trigger large transcriptional responses [50,52]. Thus, the presence of alginate breakdown products may likely result in increased expression of all components of the alginate degradation pathway, including the expression of degrading enzymes. As the gene expression analysis was performed on well-mixed cultures in culture medium containing alginate breakdown products, we therefore expect a strong stimulation of alginate catabolism. In a natural scenario, where cells disperse from a polysaccharide hotspot before its exhaustion, the expression of alginate catabolism genes may likely decrease again once the local concentration of breakdown products decreases. However, continued production of alginate lyases could also provide an advantage when encountering a new alginate source and continued production of alginate lyases may thus help cells to prepare for likely future environments. Further investigations of bacterial enzyme secretion in changing nutrient environments and at relevant spatial scales are required to improve our understanding of the regulation of enzyme secretion along nutrient gradients."

(8) I like Figure 6, and I think this hypothesis is a good result from this paper, but I think it would be important to emphasize this as a proposal that needs further quantitative analysis to be supported.

We have now edited the manuscript to make this point more clear. While both degradation and dispersal are well-appreciated parts of microbial ecology, the transitions and underlying mechanisms are unclear. We have edited the discussion to improve the clarity (line 419):

"This cycle of biomass degradation and dispersal has long been discussed in the context of foraging e.g., [44,45,13,46,47], but the cellular mechanisms that drive the cell dispersal remain unclear."

Also, we have updated Figure 6 to indicate more clearly which new findings this work proposes (now bold font) and which previous findings that were made in different bacterial taxa and carbon sources that aligns with our work (now light font). We edited the figure legend accordingly (line 503):

"By integrating our results with previous studies on cooperative growth on the same system, as well as results on dispersal cycles in other systems, we highlight where the specific results of this work add to this framework (bold font)."

Minor comments

(1) Is there any growth on the enzyme used for alginate digestion? E.g. is the enzyme used to digest the alginate at sufficiently high concentrations that cells could utilize it for a carbon/nitrogen source?

We thank the reviewer for raising this point. We added the following paragraph as Supplemental Text to address it (line 179):

"Protein amount of the alginate lyases added to create digested alginate

Based on the following calculation, we conclude that the amount of protein added to the growth medium by the addition of alginate lyases is so small that we consider it negligible. In our experiment we used 1 unit/ml of alginate lyases in a 4.5 ml solution to digest the alginate. As the commercially purchased alginate lyases are 10,000 units/g, our 4.5 ml solution contains 0.45 mg of alginate lyase protein. The digested alginate solution diluted 45x when added to culture medium. This means that we added 0.18 µg alginate lyase protein to 1 ml of culture medium.

As a comparison, for 1ml of alginate medium, 1000µg of alginate is added or for 1 ml of Lysogeny broth (LB) culture medium, 3,500 µg of LB are added. Thus, the amount of alginate lyase protein that we added is ca. 5000 - 20,000 times smaller than the amount of alginate or LB that one would add to support cell growth. Therefore, we expect the growth that the digestion of the added alginate lyases would allow to be negligible."

(2) The lines in Figure 2B are very hard to see.

We have addressed this comment by using thicker lines in Figure 2B.

(3) The black background and images in Figure 3A and B are hard to see as well.

We have now replaced Figure 3A and B, now using a white background.

(4) Typo at the beginning of line 251?

Unfortunately we failed to find the typo referred to. We are happy to address it if it still exists in the revised manuscript.

Reviewer #3 (Recommendations For The Authors):

(1) I think there is not enough experimental evidence to conclude that the underlying cause of increased motility is the accumulation of digested alginate products. To conclusively show that this is the cause and not just some signal linked to cell density, perhaps the experiment should be repeated with a different carbon source.

We thank the reviewer for their comment, which made us realize that we did not make the nature of the dispersal cue clear. The gene expression data was obtained from batch cultures and measured at the same approximate bacterial densities in batch, which indeed shows that the digested alginate is a sufficient signal for an increase in motility gene expression. This agrees very well with our observation that cells growing on digested alginate in microfluidic chambers have an increased fraction of motile cells in comparison with cells exposed to alginate (Fig 3E). However, we did not mean to suggest that the observed dispersal by bacterial motility is not influenced by cell density, in fact, we see that dispersal (and hence the increase in cell motility) in microfluidic chambers that are switched from polymeric to digested alginate depends on the bacterial density in the chamber, with higher bacterial densities showing increased dispersal. This shows that the presence of alginate oligomers does trigger dispersal through motility, but this signal affects bacterial groups in a cell density dependent manner.

Similar observations have been made in *Caulobacter crescentus*, which was found to form cell groups on the polymer xylan while cells disperse when the corresponding monomer xylose becomes available (D'Souza et al., 2021). We reference the additional work in lines 179 and 230. Taken together, these observations indicate a more general phenomenon in dispersal from polysaccharide substrates.

(2) About the expression data:

• *Ribosomal proteins and ABC transporters are enriched in cells grown on digested alginate and the authors discuss that this explains the difference in max growth rate between alginate and digested alginate. However, in Figure S2E the authors report no statistical difference between growth rates.*

We have now edited the manuscript to clarify this point. We found that cells grown on degradation products reached their maximal growth rate around 7.5 hours earlier (Fig. S2D) and showed increased expression of ribosomal biosynthesis and ABC transporters in late-exponential phase (Fig. 4A). We consider this shorter lag time as a sign of a different growth state and therefore a possible reason for the difference in ribosomal protein expression.

As the reviewer correctly points out, the maximum growth rates that were computed from the two growth curves were not significantly different (Fig. S2E). However, for our gene expression analysis, we harvested the transcriptome of cells that reached OD 0.39-0.41 (mid-to late-exponential phase). At this time point, the cell cultures may have differed in their momentary growth rate.

We edited the manuscript to make this clearer (line 287):

“Both observations likely relate to the different growth dynamics of *V. cyclitrophicus* ZF270 on digested alginate compared to alginate (Fig. S2A), where cells in digested alginate medium reached their maximal growth rate 7.5 hours earlier and thus showed a shorter lag time (Fig. S2D). As a consequence, the growth rate at the time of RNA extraction (mid-to-late exponential phase) may have differed, even though the maximum growth rate of cells grown in alginate medium and digested alginate medium were not found to be significantly different (Fig. S2E).”

• *The increased expression of transporters for lyases in cells grown on digested alginate (lines 273-274 and 325-328) is very confusing and the explanation provided in lines 412-420 is not very convincing. My two cents on this: Expression of more enzymes and induction of motility might be a strategy to be prepared for more likely future environments (after dispersal, alginate is the most likely carbon source they will find).*

This would be in line with observed increased chemotaxis towards the polymer rather than the monomer (Similar to C. elegans).

This comment is in line with reviewer 2, comment 7. In response to these two comments (and as described above), we expanded our discussion of these results to contextualize these findings (around line 443):

“One reason for this observation may be that cells primarily rely on intracellular monosaccharide levels to trigger the upregulation of genes associated with polysaccharide degradation and catabolism, as has previously been observed for *E. coli* across various carbon sources [50,51]. In fact, the majority of carbon sources are sensed by prokaryotes through one-component sensors inside the cell [50]. In the one-component internal sensing scheme, the enzymes and transporters for the use of various carbon sources are expressed at basal levels, which leads to an increase in pathway intermediates upon nutrient availability. The pathway intermediates are sensed by an internal sensor, usually a transcription factor, and lead to the upregulation of transporter and enzyme expression [50,51]. This results in a positive feedback loop, which enables small changes in substrate abundance to trigger large transcriptional responses [50,52]. Thus, the presence of alginate breakdown products may likely result in increased expression of all components of the alginate degradation pathway, including the expression of degrading enzymes. As the gene expression analysis was performed on well-mixed cultures in culture medium containing alginate breakdown products, we therefore expect a strong stimulation of alginate catabolism. In a natural scenario, where cells disperse from a polysaccharide hotspot before its exhaustion, the expression of alginate catabolism genes may likely decrease again once the local concentration of breakdown products decreases. However, continued production of alginate lyases could also provide an advantage when encountering a new alginate source and continued production of alginate lyases may thus help cells to prepare for likely future environments. Further investigations of bacterial enzyme secretion in changing nutrient environments and at relevant spatial scales are required to improve our understanding of the regulation of enzyme secretion along nutrient gradients.”

Additionally, we agree with the intriguing comment that continued expression of alginate lyases may also prepare cells for likely future environments. Further studies that aim to answer whether marine bacteria are primed by their growth on one carbon source towards faster re-initiation of degradation on a new particle will be an interesting research question. We now address this point in our manuscript (line 458):

“However, continued production of alginate lyases could also provide an advantage when encountering a new alginate source and continued production of alginate lyases may thus help cells to prepare for likely future environments. Further investigations of bacterial enzyme secretion in changing nutrient environments and at relevant spatial scales are required to improve our understanding of the regulation of enzyme secretion along nutrient gradients.”

(3) The yield reached by Vibrio on alginate is significantly higher than the yield in digested alginate, not similar, as stated in lines 133-134. Only cell counts are similar. Perhaps the author can correct this statement and speculate on the reason leading to this discrepancy: perhaps cells tend to aggregate in alginate despite the fact that these are well-mixed cultures.

We have edited the description of the OD measurements accordingly and agree with the reviewer that aggregation is indeed a possible reason for the discrepancy (line 141):

“We also observed that the optical density at stationary phase was higher when cells were grown on alginate (Fig. S2B and C). However, colony counts did not show a significant

difference in cell numbers (Fig. S3), suggesting that the increased optical density may stem from aggregation of cells in the alginate medium, as observed for other *Vibrio* species [7].”

(4) I suggest toning down the importance of the results presented in this study for understanding global carbon cycling. There is a link but at present it is too much emphasized.

We have edited our statements regarding the carbon cycle. In the revised manuscript we stress the lack of direct quantifications of carbon cycling. . We still refer to carbon flow in the revised manuscript, as we would argue that microbial remineralization of biomass is recognized as an important factor in the marine biological carbon pump (e.g., Chisholm, 2000) and research on marine bacterial foraging investigates how bacterial cells manage to find and utilize this biomass.

Our revised manuscript contains the following modified statements (line 47 and line 60): “Even though many studies indicate that these degradation-dispersal cycles contribute to the carbon flow in marine systems, we know little about how cells alternate between polysaccharide degradation and motility, and which environmental factors trigger this behavioral switch.”

“Overall, our findings reveal cellular mechanisms that might also underlie bacterial degradation-dispersal cycles, which influence the remineralization of biomass in marine environments.”

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